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The Potential of Radiomagnetotellurics for Detecting Dike Instabilities

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Abstract

Understanding the internal structure of water protection structures such as dikes and embankments is crucial for water safety management. Methods based on electrical resistivity are well suited for this purpose, as instabilities within dikes can produce distinct resistivity contrasts. In addition to the commonly used electrical resistivity tomography (ERT), this project investigates the potential of radiomagnetotellurics (RMT) for characterizing the internal structure of dikes and detecting potential instability features. Particular emphasis is placed on characterizing the RMT data, assessing data quality, evaluating the information provided by the impedance tensor and tipper function, and testing the performance of RMT inversions to establish an overall assessment of its potential for dike investigations. The study combines synthetic modeling, RMT and ERT data acquisition at four dike locations, and all stages from data processing and inversion to interpretation. This allows the suitability of RMT to be assessed across the full workflow from data acquisition to interpretation. ERT data were acquired primarily for comparison, while available geotechnical information at the different sites provided additional constraints for interpretation. The synthetic modeling demonstrates that RMT is sensitive to instability features expected within dikes, with a substantially stronger response to conductive than resistive features. The field data show a similar sensitivity contrast, even though no specific instability features could be identified with confidence. The results also highlight the quality of the high-frequency transfer functions as a key limitation for near-surface resolution. Improving the reliability of high-frequency data is therefore essential for increasing the applicability of RMT to dike investigations. Synthetic modeling further shows that dike topography strongly influences the RMT response, indicating that the data are affected not only by the internal structure of the dike but also by its geometry. The tipper data are also strongly affected by noise, dike topography, and surrounding structures, making their interpretation in terms of the internal dike structure challenging. Despite these limitations, the strong sensitivity of RMT to conductive features and its ability to provide information on their orientation highlight its potential as a complementary tool for targeted dike investigations.

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Chapter 1

Introduction

Rising sea levels, an increasing frequency of heavy rainfall and flooding, and continued population growth in flood-prone areas have increased the demand for effective flood protection. Embankments and dikes play a crucial role in mitigating flood hazards. Their failure can have severe consequences, potentially resulting in catastrophic flooding. It is therefore essential to regularly inspect and monitor these structures to identify potential vulnerabilities at an early stage. Common approaches include visual inspections, geotechnical investigations, and geophysical surveys, each with its own advantages and limitations. Visual inspections are often unable to detect internal processes such as internal erosion or piping, as these may not produce visible surface indications (Ball et al., 2023). Geotechnical methods, such as cone penetration tests, provide accurate and reliable measurements, but are destructive and only provide stratigraphic information at specific locations. Geophysical methods are non-destructive and enable the spatial characterization of subsurface properties, but can be relatively time-consuming and costly to carry out. Since dikes often extend over large distances, investigation methods should also be relatively fast. In the Netherlands alone, the flood protection system comprises 94 dike rings with a total length of approximately 3,800 km of dikes (Jorissen et al., 2016). Given these circumstances, selecting an appropriate method for dike investigations is not a trivial task. Rather than relying on a single technique, the most effective assessment is often achieved by combining complementary methods that exploit their respective strengths while compensating for their individual limitations.

A systematic dike investigation should begin with the collection and evaluation of all available information about the structure. This should be followed by rapid geophysical surveys to identify anomalous or heterogeneous zones within the dike. In the final step, these areas can be investigated in greater detail using geotechnical methods or targeted geophysical surveys (Fauchard & Mériaux, 2007). This project aims to investigate the suitability of Radiomagnetotellurics (RMT) for dike investigations, both as a rapid screening method and as a tool for detailed subsurface characterization. Detailed investigations could be conducted using a dense grid of RMT stations, where both electric and magnetic field components are measured to provide high-resolution information on the subsurface. For rapid screening, RMT could potentially be adapted to drone-based surveys using measurements of the magnetic field components only, enabling efficient coverage of larger areas.

The suitability of electromagnetic methods for dike investigations is well established, as instabilities such as cavities or water flow paths show a contrast in electrical resistivity compared to the rest of the dike (Niederleithinger et al., 2013). The most commonly applied method is Electrical Resistivity Tomography (ERT), which has demonstrated promising results in numerous studies (Niederleithinger et al., 2013). However, setting up ERT lines is time consuming and measuring large extents of dikes with ERT is not feasible. Also, ERT is typically measured along a line in longitudinal direction of the dike. Recent studies have shown that such measurements can be significantly affected by three-dimensional effects arising from factors such as dike topography and adjacent water bodies (Ball et al., 2023; Fargier et al., 2014). Diffusive electromagnetic methods such as RMT are generally more sensitive to conductive features, whereas ERT is typically more sensitive to resistive structures (Candansayar & Tezkan, 2008; Özyıldırım et al., 2020). RMT may therefore be particularly suitable for detecting conductive anomalies. Furthermore, combining RMT with ERT has the potential to provide a more complete characterization of the subsurface by exploiting the complementary sensitivities of the two methods. RMT has already been demonstrated to be applicable for environmental investigations (Rulff et al., 2025; Tezkan, 1999). However, dike studies introduce additional complexity due to the complex topography and varying hydraulic conditions associated with these structures.

In this thesis I assess the applicability of RMT under these challenging conditions. Notably, I assess the suitability of RMT for dike investigations by considering the complete workflow from synthetic modeling and field acquisition to data processing, inversion, and interpretation. This allows me to evaluate the method at each step and identify the factors that ultimately affect the reliability of the results. In particular, I investigate the effects of the three-dimensional geometry and topography of the dikes on the RMT data and assess what information can be extracted from data that are affected by substantial noise. Furthermore, the availability of complementary information, including ERT, CPT, and TEM data, allows me to assess which features identified in the RMT models are supported by independent observations and which cannot be reliably resolved. This provides a basis for evaluating the potential of RMT as a tool for investigating the internal structure of dikes and identifying the main limitations of the method. I found that RMT is applicable in dike environments, but that several factors need to be carefully considered to extract reliable information from the data, including accurate representation of topography, careful data processing, and assessing the reliability of different frequency ranges before including them in the inversion.

After introducing the principles of RMT and briefly discussing the types of instability features expected within dikes in Chapter 2, I first present a synthetic study to assess the sensitivity of RMT and the detectability of such features. In Chapter 3, I then introduce two of the four dike locations where RMT data were acquired and investigate these two sites in detail. The remaining two sites are described in the Appendix A. For the two selected sites, I discuss the data processing and resulting data quality in Chapter 4, followed by the inversion of the RMT data and the interpretation of the resulting resistivity models in Chapter 5. I also investigate the effects of the three-dimensional dike topography on the RMT data and discuss the implications for the inversion results. Finally, I bring the results together to assess the potential and limitations of RMT for dike investigations.

Chapter 2

Method

The methods applied in this project are Radiomagnetotellurics (RMT) and Electrical Resistivity Tomography (ERT). Although they are based on different principles, both methods aim to determine the subsurface distribution of electrical resistivity. In the context of dike investigations, electrical resistivity is a useful parameter because it can be used as a proxy for properties such as water content, porosity, and groundwater salinity (Cheng et al., 2025).

The physics behind both methods is governed by Maxwell's equations. However, there are clear differences between them. In ERT, a stationary electric potential field is measured, which simplifies the underlying physics. As a direct current (DC) method, ERT investigates the distribution of the electric potential in the subsurface. In RMT, time-dependent electromagnetic fields are measured. These fields diffuse into the subsurface according to the diffusion equation, and the measured response depends on both the electrical resistivity of the subsurface and the measurement frequency. Because of the increased complexity, RMT data require significantly more processing before they can be inverted, whereas the output of ERT, the apparent resistivity, can be directly used for inversion. For both methods, inversion is used to obtain a model of the subsurface that explains the measured data within their associated uncertainties.

2.1 Radiomagnetotellurics

The basic concept of Radiomagnetotellurics (RMT) is closely related to that of magnetotellurics (MT), with the main difference being the used frequency range. RMT operates in the radio frequency band, typically between 10 kHz and 1 MHz, and uses electromagnetic signals from existing radio transmitters as the source field. RMT is a frequency-domain electromagnetic method based on the principle that time-varying electromagnetic fields induce electrical currents in the subsurface. These induced currents generate secondary electromagnetic fields that contain information about the electrical resistivity distribution of the subsurface. To characterize these fields, the two horizontal components of the electric field and the three components of the magnetic field are measured using grounded electrodes and a magnetometer, respectively. A major advantage of RMT is that no dedicated transmitter is required for the survey. Instead, existing radio transmitters provide the source field, which simplifies field logistics and reduces the equipment required for data acquisition. The propagation of electromagnetic fields in the subsurface is described by Maxwell's equations. The relevant processes are mainly controlled by the interaction between changing magnetic fields and induced electrical currents. These processes are described by Faraday's law and Ampère's law (Ward & Hohmann, 1987),

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \ , \tag{2.1}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \ , \tag{2.2}$$

where $\mathbf{E}$ is the electric field (V/m), $\mathbf{H}$ is the magnetic field intensity (A/m), $\mathbf{B}$ is the magnetic induction (T), $\mathbf{J}$ is the conduction current density (A/m^2), and $\mathbf{D}$ is the electric displacement field (C/m^2).

These equations can be simplified by applying Ohm's law, $\mathbf{J} = \sigma\mathbf{E}$, where σ is the electrical conductivity (S/m), together with the constitutive relations $\mathbf{H} = \mathbf{B}/\mu$ and $\mathbf{D} = \epsilon\mathbf{E}$. The inverse of the electrical conductivity is the electrical resistivity, ρ ($\Omega\,$m), which is used throughout most of this text. The first assumption is that both the electrical permittivity, ϵ (F/m), and the magnetic permeability, μ (H/m), are spatially constant and can be approximated by their free-space values, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 = 1.2566 \times 10^{-6}$ H/m, respectively (Simpson & Bahr, 2005). Substituting these relations into Equations 2.1 and 2.2, and transforming the equations into the frequency domain, yields:

$$\nabla \times \mathbf{E} = -i\omega\mathbf{B} \tag{2.3}$$

$$\nabla \times \mathbf{B} = \mu_0\sigma\mathbf{E} + i\epsilon_0\mu_0\omega\mathbf{E} \tag{2.4}$$

Here, ω (rad/s) represents the angular frequency. Taking the curl of these equations yields the following expression:

$$\nabla^2 F = (i\omega\mu_0\sigma - \omega^2\mu_0\epsilon_0)F \tag{2.5}$$

where F can represent either the electric field $\mathbf{E}$ or the magnetic field $\mathbf{B}$. The right-hand side of this equation consists of two terms, where the first is related to conduction current and the second to displacement current (Tezkan, 2009). Equation (2.5) can be further simplified by neglecting the second term, provided that the condition $\sigma \gg \omega\epsilon_0$ is satisfied. This assumption is commonly made in magnetotellurics, where the displacement current is negligible compared to the conduction current. For RMT, however, the higher operating frequencies require verification that this condition remains valid. Using the highest frequency considered in this study, $\omega = 2\pi \times 10^6$ rad/s, together with the free-space permittivity, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m, gives $\omega\epsilon_0 \approx 5.6 \times 10^{-5}$ S/m. This value is several orders of magnitude smaller than the expected conductivity of the dike (approximately 0.2 S/m). Consequently, the displacement current can be neglected, and Equation (2.5) reduces to the quasi-static approximation (Simpson & Bahr, 2005). The resulting equation is a diffusion equation. Unlike electromagnetic waves propagating in free space, where wave propagation dominates, MT and RMT fields in the Earth are primarily governed by diffusive processes and are therefore attenuated as they penetrate the subsurface. Lower frequencies can penetrate to greater depths. Consequently, measurements at different frequencies provide information about the electrical conductivity structure at different depths. A useful quantity to evaluate how deep electromagnetic signals penetrate is the skin depth, which tells us at what depth the signal has been attenuated by 1/e (Tezkan, 2009). The skin depth, expressed in meters, can be approximated as

$$\delta = 503\sqrt{\rho/f} \quad . \tag{2.6}$$

At sufficient distances from the source and by considering contributions from transmitters in all directions, the incoming electromagnetic field can be approximated as a plane wave (McNeill & Labson, 1991; Zacher, 1992). Under this assumption, the primary electromagnetic field is considered to be horizontally polarized, as the wave is assumed to propagate vertically towards the Earth's surface, resulting in electric and magnetic field components that are predominantly horizontal. The electromagnetic system can be described as a linear relationship between the horizontal components of the magnetic field and the corresponding horizontal components of the electric field. The Earth response function relating these quantities is known as the impedance tensor and is expressed as

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} \begin{bmatrix} H_x \\ H_y \end{bmatrix} \quad . \tag{2.7}$$

For a 1D Earth, the diagonal elements of the impedance tensor are zero, while the off-diagonal elements have equal magnitudes and opposite signs (Simpson & Bahr, 2005). For a 2D subsurface, the tensor can be rotated into the geoelectric strike direction, resulting in vanishing diagonal elements. In a 3D subsurface, all four impedance tensor elements can be non-zero (Simpson & Bahr, 2005). The apparent resistivities, ρ_a, and phases, ϕ, are calculated from the impedance tensor elements as

$$\rho_{a,ij} = \frac{1}{\mu_0 \omega} |Z_{ij}|^2 \ , \quad \text{and} \quad \phi_{ij} = \arctan\left(\frac{\Im(Z_{ij})}{\Re(Z_{ij})}\right) \ . \tag{2.8}$$

Here, $\Re(Z_{ij})$ and $\Im(Z_{ij})$ denote the real and imaginary parts of the impedance tensor element Z_{ij}, respectively. Further information about subsurface conductivity variations can be obtained from the tipper function, which describes the relationship between the vertical and horizontal components of the magnetic field:

$$H_z = T_{xz} H_x + T_{yz} H_y \ , \tag{2.9}$$

where H_x, H_y, and H_z denote the magnetic field components in the x, y, and z directions, respectively. The quantities T_{xz} and T_{yz} are the complex tipper components that describe the relation between the horizontal and vertical magnetic fields (Naidu, 2012). The impedance tensor and tipper function are commonly referred to collectively as transfer functions. In practice, the impedance tensor is estimated from the cross- and auto-spectra of the measured electromagnetic field components. The individual tensor elements can be calculated with (Weckmann, 2025)

$$
\begin{aligned}
Z_{xx} &= \frac{[H_y^* H_y][H_x^* E_x] - [H_y^* H_x][H_y^* E_x]}{[H_x^* H_x][H_y^* H_y] - [H_x^* H_y][H_y^* H_x]} \ , \\
Z_{xy} &= \frac{[H_y^* H_y][H_x^* E_y] - [H_y^* H_x][H_y^* E_y]}{[H_x^* H_x][H_y^* H_y] - [H_x^* H_y][H_y^* H_x]} \ , \\
Z_{yx} &= \frac{[H_y^* H_x][H_x^* E_x] - [H_y^* H_y][H_x^* E_x]}{[H_x^* H_x][H_y^* H_y] - [H_x^* H_y][H_y^* H_x]} \ , \\
Z_{yy} &= \frac{[H_x^* H_x][H_y^* E_y] - [H_x^* H_y][H_x^* E_y]}{[H_x^* H_x][H_y^* H_y] - [H_x^* H_y][H_y^* H_x]} \ .
\end{aligned}
\tag{2.10}
$$

2.2 Electrical Resistivity Tomography

Electrical Resistivity Tomography (ERT) is not the main focus of this project, and therefore only a brief introduction to the method is provided here. In ERT surveys, electrodes are placed along a profile with constant spacing and are used to inject electrical current into the subsurface and measure the resulting potential differences. Each measurement involves four electrodes: two electrodes act as the current source and sink, while the remaining two measure the voltage difference between them. By knowing the injected current, the measured voltage, and the geometric factor of the electrode configuration, the apparent resistivity of the subsurface can be calculated using Ohm's law (Rubin & Hubbard, 2005). The apparent resistivity represents the resistivity of a homogeneous half-space that would produce the same measured voltage for the given electrode configuration and injected current. By repeating measurements using different combinations of transmitting and measuring electrode pairs along the profile, a large number of apparent resistivity values are obtained. These values represent different positions and apparent depths within the subsurface. The resulting measurements are commonly visualized as a pseudosection, which serves as input data for an inversion procedure to estimate a model of the true electrical resistivity distribution of the subsurface.

2.3 Dike Structures and Instabilities

Dikes are water-retaining structures constructed from unconsolidated materials, primarily sand, peat, and clay. Historically, they were built using locally available materials. Older dikes often exhibit a layered composition of clay and peat overlying a sand foundation (Olalla, 2024). Modern dikes, in contrast, are generally constructed with a sand core and a clay cover layer, which reduces permeability and improves water retention (Jonkman, 2014). In the Netherlands, where flood protection has been an essential part of land management for centuries, many dikes are several hundred years old and have undergone numerous reinforcements and modifications. Rather than being completely rebuilt, these reinforcements often consisted of adding new material on top of or alongside the existing structure. Most dikes remain dry under normal, non-flood conditions, meaning that there is typically no water within the dike body (Fauchard & Mériaux, 2007).

Dikes are critical infrastructure, as they protect populated areas and other valuable infrastructure from flooding. Regular monitoring is therefore necessary to assess their structural integrity and identify potential failures at an early stage, allowing preventive measures to be taken. Dike failure can generally be categorized into four main failure mechanisms: overtopping, surface erosion, internal erosion, and slope failure (Dunbar et al., 2017). Overtopping occurs when water levels rise above the crest elevation of the dike, causing water to flow over the structure. Surface erosion results from the scouring action of flowing water along the dike surface, typically during high-water events. Internal erosion is caused by seepage through the dike body, which can mobilize and remove soil particles, potentially leading to the formation of internal channels and structural weakening. Slope failure involves the loss of stability of the dike slopes and may be limited to the dike body itself or extend into the underlying foundation material (Dunbar et al., 2017). Instability features within dikes can develop in the form of macro pores, such as cracks resulting from differential settlement or shrinkage–swelling processes (Borgatti, 2017), as well as voids caused by animal burrows or the decay of tree roots (Fauchard & Mériaux, 2007). These weaknesses create preferential flow paths through the dike body, which can facilitate internal erosion and eventually lead to piping. Piping occurs when seepage flow initiates the detachment and transport of soil particles, gradually forming an internal channel within the dike. This process typically initiates on the landward side of the dike, commonly referred to as the passive side, and progresses towards the waterside, or active side Piping is recognized as one of the primary causes of hydraulic structure failure and is responsible for approximately 50 % of dam failures worldwide (Borgatti,

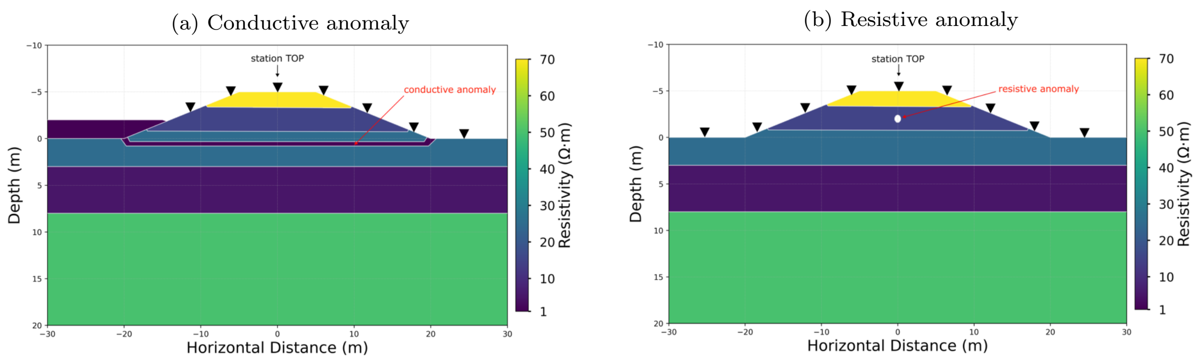


Figure 2.1: Synthetic 2D models representing in (a) a piping case, and in (b) an animal burrow. The models were created using MARE2DEM (Key, 2016). Station TOP indicates the station for which the response is shown in Figure 2.2.

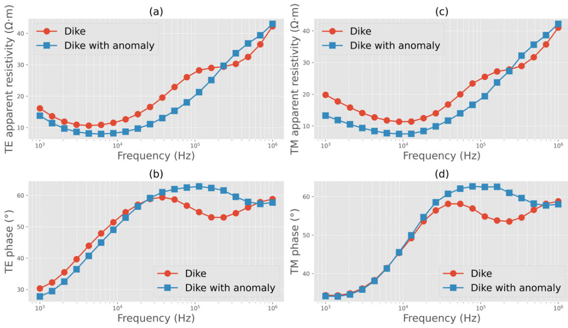
2017). In the Netherlands, animal burrows are a common cause of void formation within dikes. These burrows are typically created by species such as beavers, foxes, rabbits, and moles (Tsimopoulou & Koelewijn, 2023).

2.4 2D Modeling of Dike instabilities

The first step of this project was a synthetic modeling study to assess the applicability of RMT for dike investigations. The objective was to evaluate the sensitivity of RMT to potential anomalies within the dike and to investigate whether responses associated with instability features can be detected above the background noise level. The modeling was performed with MARE2DEM (Key, 2016) in a two-dimensional framework, primarily to reduce complexity and runtime. For the baseline model, a layered dike structure was constructed using resistivity values derived from the ERT inversion at one of the measurement sites. Two separate models were then developed, each incorporating a different type of instability feature: one representing a void within the dike and the other simulating piping effects. The resulting models are shown in Figures 2.1a and 2.1b.

With these models, forward responses were calculated and compared to the responses given by the baseline model. This is displayed in Figure 2.2. The conductive anomaly (Figure 2.1a) produces a much more pronounced response than the resistive anomaly (Figure 2.1b), making it considerably easier to distinguish from the baseline model. This is probably caused by the higher sensitivity of RMT to conductive features compared to resistive features. Additionally, the resistive anomaly can be modeled more accurately in 2D. In a 2D model, the piping anomaly is effectively represented as a thin, infinitely

(a) Conductive anomaly



(b) Resistive anomaly

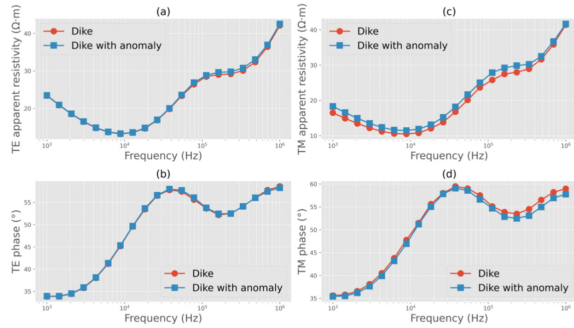
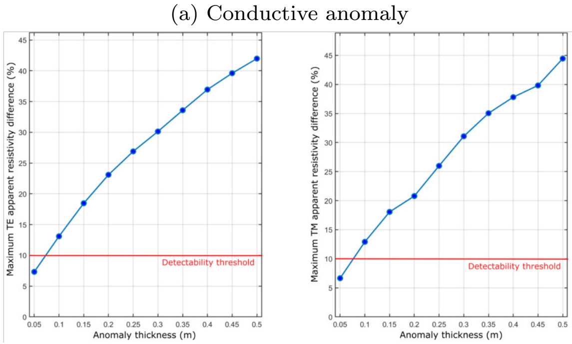


Figure 2.2: Comparison of the TE and TM responses at station TOP, for the models with and without the anomaly; (a) conductive anomaly, (b) resistive anomaly.

(a) Conductive anomaly



(b) Resistive anomaly

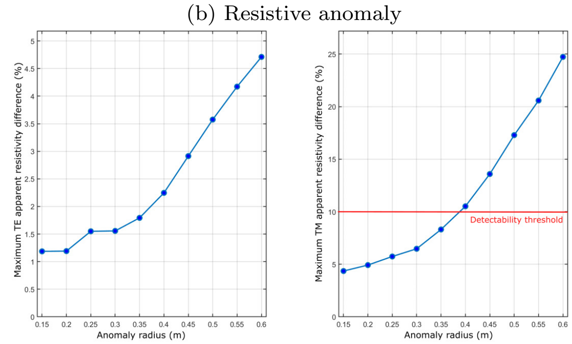


Figure 2.3: Maximum recorded difference between the model containing an instability feature and the corresponding baseline model. (a) Piping layer representing a conductive anomaly with thicknesses ranging from 0.05 to 0.5 m for the TE and TM modes. (b) Void representing a resistive anomaly with radii ranging from 0.15 to 0.6 m.

extended conductive layer, whereas a real piping feature is unlikely to extend very far along the longitudinal direction of the dike. The response of the conductive anomaly is therefore likely overestimated in the 2D model. A 3D approach would therefore be required to more realistically represent the finite extent of instability features and their associated RMT response. Nevertheless, this demonstrates that RMT is likely capable of detecting the effects of piping.

The thickness of the instability features was then varied to investigate the minimum size at which they could potentially be detected. A difference of 10 % between the baseline response and the response containing the anomaly was assumed to represent the detection threshold, as smaller differences would be difficult to distinguish from the noise. Figure 2.3 shows the resulting differences for the conductive and resistive anomalies. For the piping layer, the results indicate that the anomaly should be detectable for thicknesses of approximately 10 cm or greater. For the void, the difference between the baseline response and the response containing the anomaly remains below 5 % for the TE mode across all tested radii. For the TM mode, the difference exceeds 10 % only for voids with radii larger than approximately 40 cm. Such a large void is unlikely to occur in a dike. However, smaller voids, such as animal burrows, may still be detectable if they form interconnected networks of paths that collectively produce a larger resistive zone.

These results indicate that conductive instability features are more favorable targets for RMT than isolated resistive features. The latter may only become detectable when multiple voids form a larger resistive structure. It should also be noted that the forward responses were calculated over a frequency range from 1 kHz to 1 MHz, which is typically considered the bandwidth of RMT. During the field measurements conducted in this study, however, the highest recorded frequencies were approximately 250 kHz.

Chapter 3

Fieldwork

The fieldwork that provided the data for this thesis was carried out in the area around Rotterdam/Delft in the Netherlands. It was gathered during 3.5 days with a team of six to seven people. As the goal of the project is the determination of the suitability of RMT for dike applications in general, the sites where chosen so that different structures could be targeted under various circumstances. This allows a general assessment of the method. For that purpose fieldwork was performed at four different dikes. The dike locations investigated during the field campaign were selected in collaboration with representatives of the regional water board, who provided guidance in identifying suitable study sites. In addition to the two RMT systems (ADU-08e, Metronix), an Electrical Resistivity Tomography (ERT) system (IRIS Instruments) was deployed. The ERT data were acquired to provide an independent geophysical dataset, enabling a direct comparison between the two methods. Furthermore, the combined use of RMT and ERT allows the potential benefits of integrated evaluation of the two approaches. Of the four dike locations measured during the field campaign, only Dike 1 and Dike 3 are discussed in detail in this thesis. Further information regarding the remaining dike locations is provided in Appendix A.

3.1 Beerenpolderse Kade - Dike 1

Dike 1 was selected because water with a salinity similar to that of the river on the active side was found on the passive side. This suggests that water may be flowing through the dike from the active to the passive side. Consequently, it represents a suitable site for investigating the capability of RMT to detect such potentially conductive features. In addition to the RMT measurements, an ERT profile was acquired along the crest of the dike. Furthermore, several Cone Penetration Test (CPT) datasets from

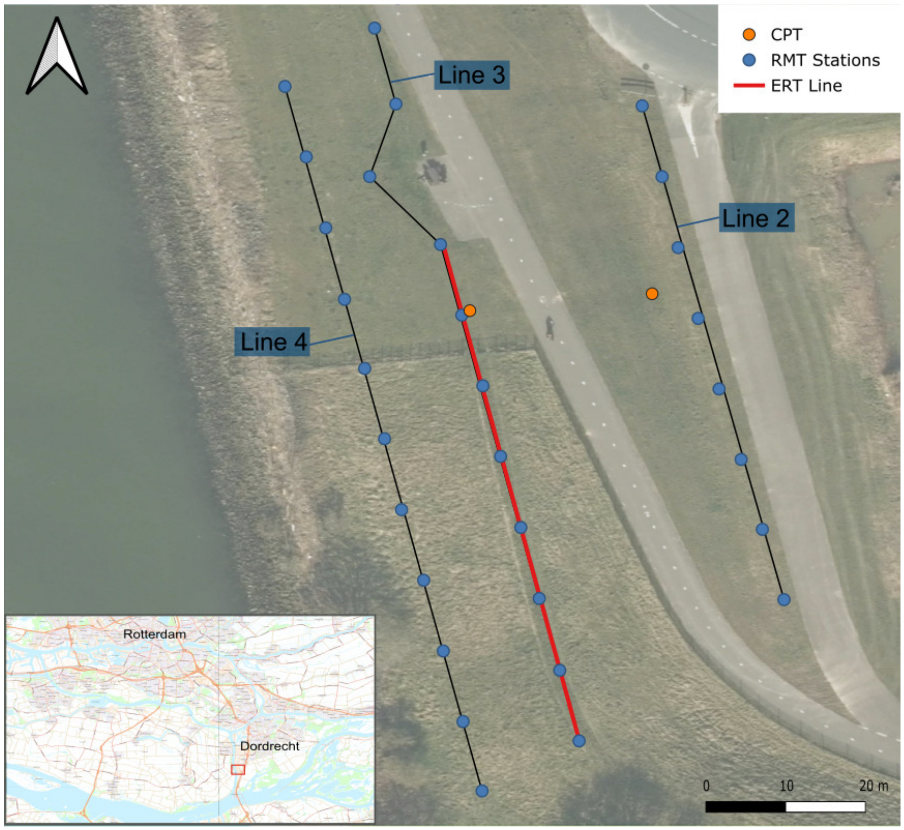


Figure 3.1: Location of the Beerenpolderse Kade (inset) and the survey layout of RMT Lines 2, 3, and 4 and one ERT line. The two marked CPT locations indicate existing CPT data that are publicly available.

the area were available for comparison and interpretation. Figure 3.1 shows the survey layout of Dike 1. The RMT profiles are referred to as Lines 2, 3, and 4, following the original survey plan in which four measurement lines were planned. As established in the synthetic modeling study in Section 2.4, RMT data acquired in dike environments must be analyzed in three dimensions to adequately account for the complex geometry of the structure. Consequently, the survey design was developed to enable a three-dimensional interpretation of the data. The RMT stations were arranged in profiles running parallel to the dike. One profile was located along the crest of the dike, while two additional profiles were positioned on either side of the embankment. This configuration resulted in a grid of RMT stations covering a substantial portion of the dike and providing the spatial coverage required for 3D analysis. An ERT section was measured covering most of Line 3.

Table 3.1 summarizes the key acquisition parameters of the field campaign. Due to incorrect calibration settings on one of the ADU-08e systems, several stations did not yield reliable data and were remeasured on the following day.

Table 3.1: Overview of acquisition parameters for Dike 1.

Parameter	**Value**
Location	51°45’3.99”N, 4°37’59.17”E
Date	09–10 March 2026
Number of measurements	29 RMT stations; 1 ERT profile
ERT Electrode spacing	1 m
RMT Electrode spacing	10 m
RMT sampling frequency	524,288 Hz
Weather conditions	Day 1: sunny; Day 2: no precipitation, but wet ground conditions due to rainfall during the morning
Instruments	2 × ADU-08e (Metronix), 1 × ERT system (IRIS Instruments)
Measurement sequence	RMT lines 3 and 4, RMT line 2, ERT line 3

3.2 Ostmolendijk - Dike 3

The Ostmolendijk represents the most complex structure investigated in this study. In contrast to the other dike locations, where a predominantly layered subsurface is expected, the subsurface at the Ostmolendijk exhibits both vertical and lateral heterogeneity. On the passive side of the dike, additional sand has been placed to enhance stability. The resulting increase in load may have induced deformation in the underlying layers, making this area of particular interest. It is known that several cracks developed and were subsequently filled with swelling clay (T. de Gast, personal communication, 2026). The added sand and a general cross-section of the dike can be seen in Figure 3.4. Furthermore, an infilled graben structure is present perpendicular to the strike of the dike, as shown in Figure 3.3. The primary objective at this site is to assess whether such lateral heterogeneities can be detected and characterized. Due to the presence of a main road on the dike crest, RMT measurements could not be conducted on top of the dike. Measurements were therefore limited to the passive side, which constituted the main area of interest. This enabled the deployment of a dense station grid while maximizing coverage of the target area. Figure 3.2 shows the survey layout. The profiles measured were with 140 m the longest lines of this campaign. Details about the acquisition parameters are provided in Table 3.2.

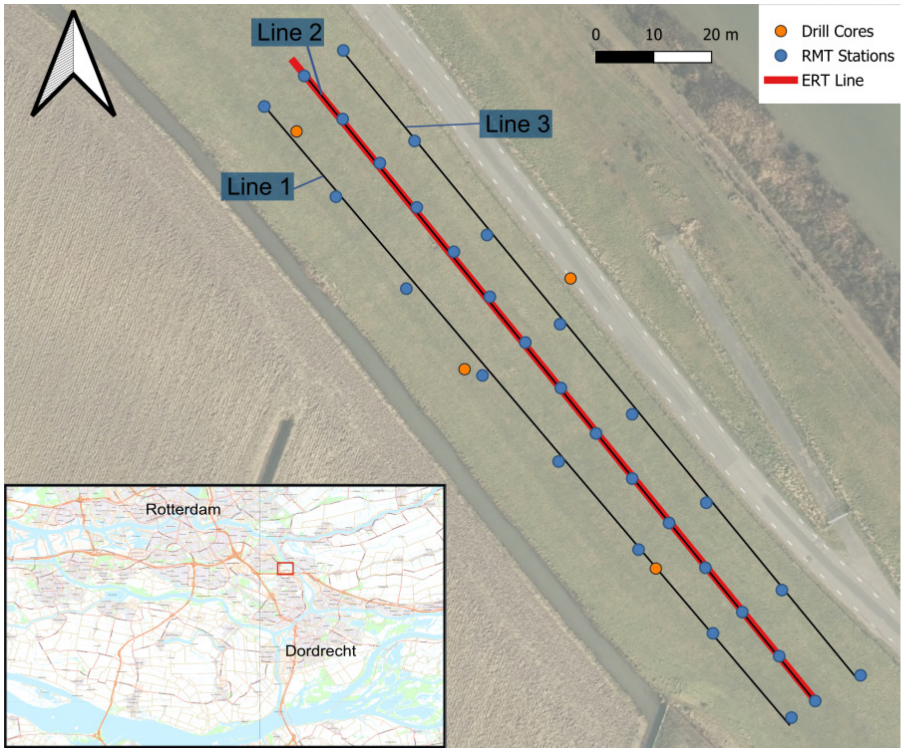


Figure 3.2: Survey layout of the Ostmolendijk (Dike 3). The location of the site is shown in the inset map. The map shows the three RMT lines, the ERT line, and the locations of existing drill cores for which publicly available data are available.

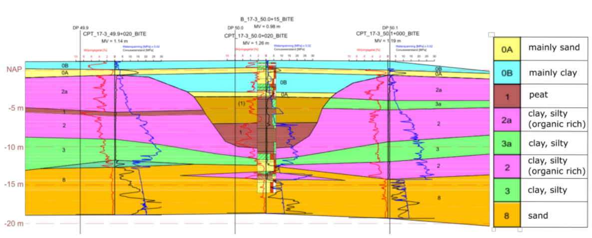


Figure 3.3: Profile along Line 2 showing a graben-like structure. The model was provided by Tom de Gast (Waterschap Hollandse Delta).

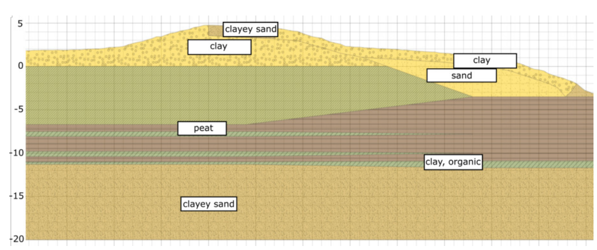


Figure 3.4: Schematic cross-section of the subsurface structure of Dike 3 perpendicular to the dike axis. The model was provided by Tom de Gast (Waterschap Hollandse Delta).

Table 3.2: Overview of acquisition parameters for Dike 3.

Parameter	Value
Location	51°51'45.64"N, 4°38'4.90"E
Date	11 March 2026
Number of measurements	31 RMT stations; 1 ERT profile
ERT Electrode spacing	2 m
RMT Electrode spacing	10 m
RMT sampling frequency	524,288 Hz
Weather conditions	windy, in the morning rain
Instruments	2 × ADU-08e (Metronix), 1 × ERT system (IRIS Instruments)
Measurement sequence	ERT line 2, RMT line 2, RMT lines 1 and 3

3.3 Considerations for RMT Surveys on Dikes

As RMT is not yet routinely applied in dike investigations, this section summarizes practical considerations and lessons learned from the field campaigns.

A key advantage of ERT compared to RMT lies in the ability to monitor data quality during acquisition. Most modern ERT systems provide real-time visualization of the developing pseudosection, allowing systematic errors, faulty electrodes, or incorrect acquisition settings to be identified and corrected directly in the field.

In contrast, RMT data processing requires several intermediate steps before apparent resistivity can be computed. As a result, equivalent real-time quality control is not possible during acquisition. This limitation places greater emphasis on careful setup, instrument checks, and field procedures prior to and during data collection. A useful first step is to inspect the amplitude spectra of the first measurement. This allows a rapid assessment of whether sufficient signal strength is present within the targeted frequency range. By comparing the spectra across all channels, it is possible to verify that each channel is recording correctly and that no technical issues are present.

In the case of using two RMT systems simultaneously, an additional consistency check can be performed by comparing the amplitude spectra of both instruments. This comparison helps to ensure that calibration, system settings, and overall instrument response are consistent and functioning as expected. However, while inspection of the amplitude spectra may be sufficient for experienced practitioners, assessing data quality based on this information alone can be challenging. If feasible, it is therefore advisable to perform a preliminary field processing of the data in order to estimate impedance values. This provides a more robust basis for evaluating data quality and identifying potential issues at an early stage.

RMT requires sufficient electrode spacing, which can be challenging in dike environments where available space is often limited. Dikes are frequently located in urban settings and are used for infrastructure as roads or bicycle paths. As a result, electrode placement may be restricted due to asphalted surfaces or other surface constraints. To address these limitations, several adjustments are possible like reducing electrode spacing, or rotating the electrode configuration. Although rotation of the setup is technically feasible, it was not applied in this study, as it increases processing complexity and requires individual rotation of the data for each station. In some cases, electrode spacing was slightly reduced. This parameter can be directly accounted for in the acquisition software and does not require additional correction during processing. Since the data are evaluated in three dimensions, individual stations can also be laterally shifted to avoid obstacles such as roads. While survey lines provide a convenient field layout, station positions can therefore be offset as needed without compromising the 3D interpretation framework. GPS coordinates are recorded at each station to ensure accurate positioning. Nevertheless, organizing stations along predefined lines is advantageous compared to a fully irregular distribution. This approach ensures a more uniform spatial coverage and facilitates a consistent station naming convention (e.g., station 5 on line 2). It also simplifies field coordination among team members during data acquisition.

Chapter 4

Data Processing

Processing the RMT data is a necessary step to derive impedance tensors, or apparent resistivity and phase values, from the five recorded time series representing the temporal variations of the electric and magnetic field components. During this process, the large time-series dataset is condensed into a frequency-domain representation consisting of the four complex impedance tensor elements and the two complex components of the tipper for each frequency. The data processing was performed with the EMERALD processing software (Ritter et al., 2015). Once the time series are transformed into the frequency domain, the data exhibit distinct peaks at frequencies corresponding to radio transmitter signals. An example of the five components of a single measurement is shown in Figure 4.1. The signal around 100 kHz is very broadband and could not effectively be used for the processing. It can be observed that all components look very similar and exhibit peaks at the same frequencies. In RMT, these spectral peaks are grouped into frequency bins, for which impedance tensors are subsequently estimated. Each frequency bin is expected to contain contributions from multiple transmitters located in different directions, thereby supporting the validity of the plane-wave assumption. Figure 4.2 shows the selected frequency file, which consists of logarithmically spaced frequency bins with 7.5 bins per decade. Logarithmic spacing is used to ensure that the relative error in the penetration depth remains approximately constant across the frequency range (Simpson & Bahr, 2005). The choice of the number of bins represents a trade-off. Using a larger number of bins increases the frequency resolution and, consequently, the amount of information that can be extracted. However, narrower bins contain contributions from fewer transmitter frequencies and directions, which can reduce the robustness of the impedance estimates. Conversely, wider bins include signals from more transmitters and a greater range of propagation directions, resulting in more accurate transfer functions. A frequency resolution of 6 to 10 bins per decade is commonly used in standard MT

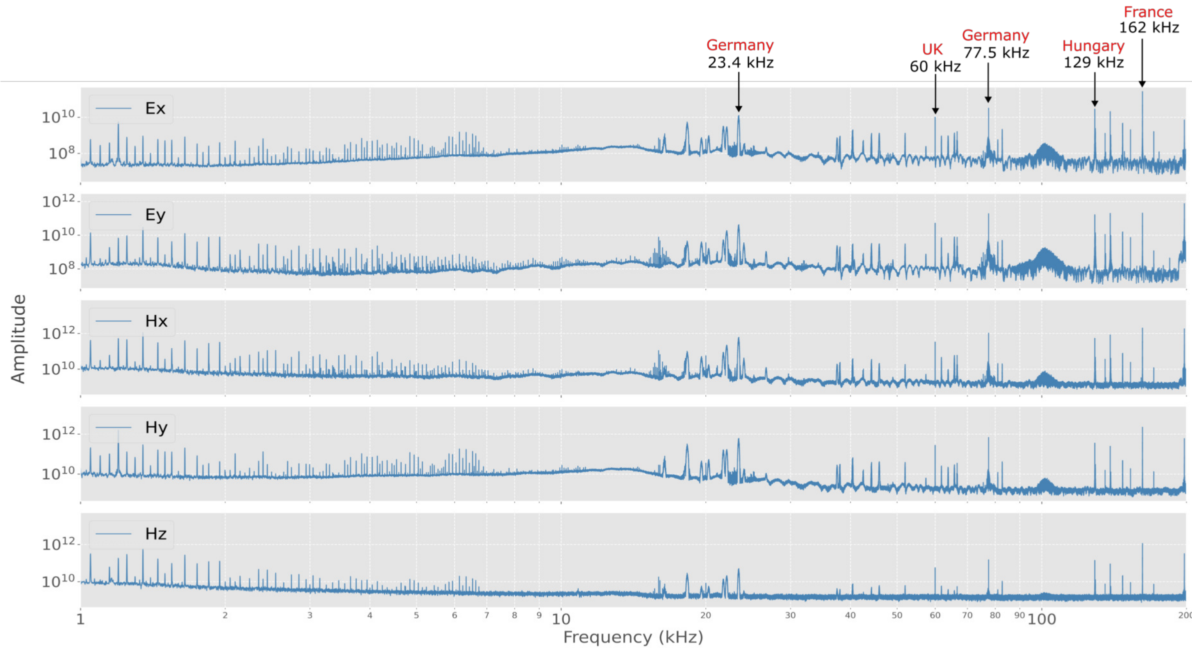


Figure 4.1: Amplitude spectra of the five components E_x, E_y, H_x, H_y, and H_z for an example station at Dike Beaver (see Appendix). The most prominent transmitter frequencies and their countries of origin are indicated.

processing (Simpson & Bahr, 2005). As RMT operates according to the same principles, a value of 7.5 bins per decade was used, which falls within this commonly used range. To increase the robustness of the transfer functions, the time series are divided into short segments, and an impedance tensor is estimated at each frequency for each segment. This produces multiple independent impedance estimates, which can then be evaluated statistically to identify outliers and quantify the uncertainty of the final estimate.

The impedance tensor describes the linear relationship between the magnetic and electric fields. It can be viewed as a linear system in which the magnetic field is the input, the

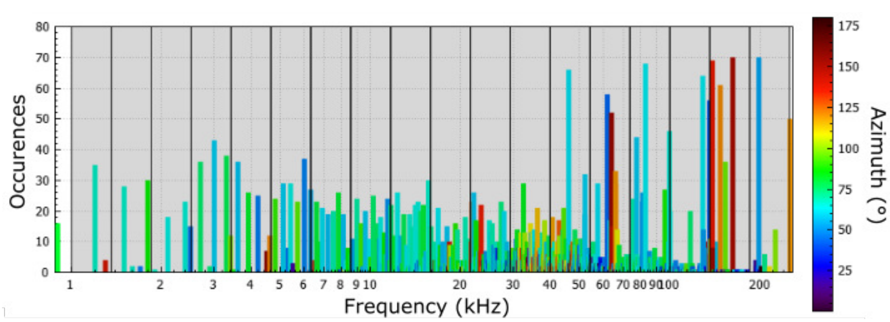


Figure 4.2: Frequency bins and the signals contained within each bin for all stations at Dike 1. The color indicates the direction of arrival of the signal. Occurrence indicates the number of time segments in which the signal at a given frequency was stronger than the noise level. This figure was created with the software rmtfreq developed at GFZ Potsdam.

impedance tensor represents the Earth's response, and the electric field is the output. Consequently, estimating the impedance tensor can be formulated as an optimization problem, in which the objective is to minimize the misfit between the observed electric field and the electric field predicted from the measured magnetic field. There are several approaches to solving this optimization problem. The most straightforward approach is to minimize the misfit in a least-squares sense. This method performs well when the data have low noise levels or when the noise is Gaussian distributed (Simpson & Bahr, 2005). However, these assumptions are often violated for RMT data, making more robust estimation methods necessary. Instead of using the L2 norm to quantify the misfit, the Mahalanobis distance is used (Platz & Weckmann, 2019), as it accounts for the covariance structure of the data and is therefore better suited to non-Gaussian distributions. In addition, a robust weighting scheme is applied, which reduces the influence of outliers on the estimated impedance tensor. For most measurements and frequencies, the coherency is sufficiently high, and the distribution of the impedance estimates is approximately Gaussian. An example of a high-quality dataset is shown in Figure 4.3a. For all impedance tensor components, the individual estimates cluster closely around the same values, indicating a stable and reliable estimate. At some stations, however, additional noise is present, most likely originating from the instrument. In these cases, a secondary cluster of impedance estimates can be observed. These estimates are typically associated with low coherency and would bias the final impedance estimate if retained. To mitigate this effect, a coherence threshold of 0.8 is applied, such that only impedance estimates with a coherency greater than 0.8 are included in the final analysis. An example of such a noisy measurement is shown in Figure 4.3b. In some cases, the impedance estimates with the highest coherency are clear outliers. A coherency criterion alone is therefore not sufficient, as high coherency can result from coherent noise rather than the desired geophysical signal. These outliers are identified and removed during the robust processing, preventing them from biasing the final impedance estimates. In addition to the robust processing and the coherence criterion, a Mahalanobis distance threshold of 1.4 was applied to further identify and reject outlying estimates. The coherence threshold and Mahalanobis distance were determined through extensive testing on individual stations by evaluating a range of values and examining the resulting impedance tensor estimates for physical meaningfulness and continuity. The selected values were subsequently applied uniformly to all stations to ensure consistent processing strategy.

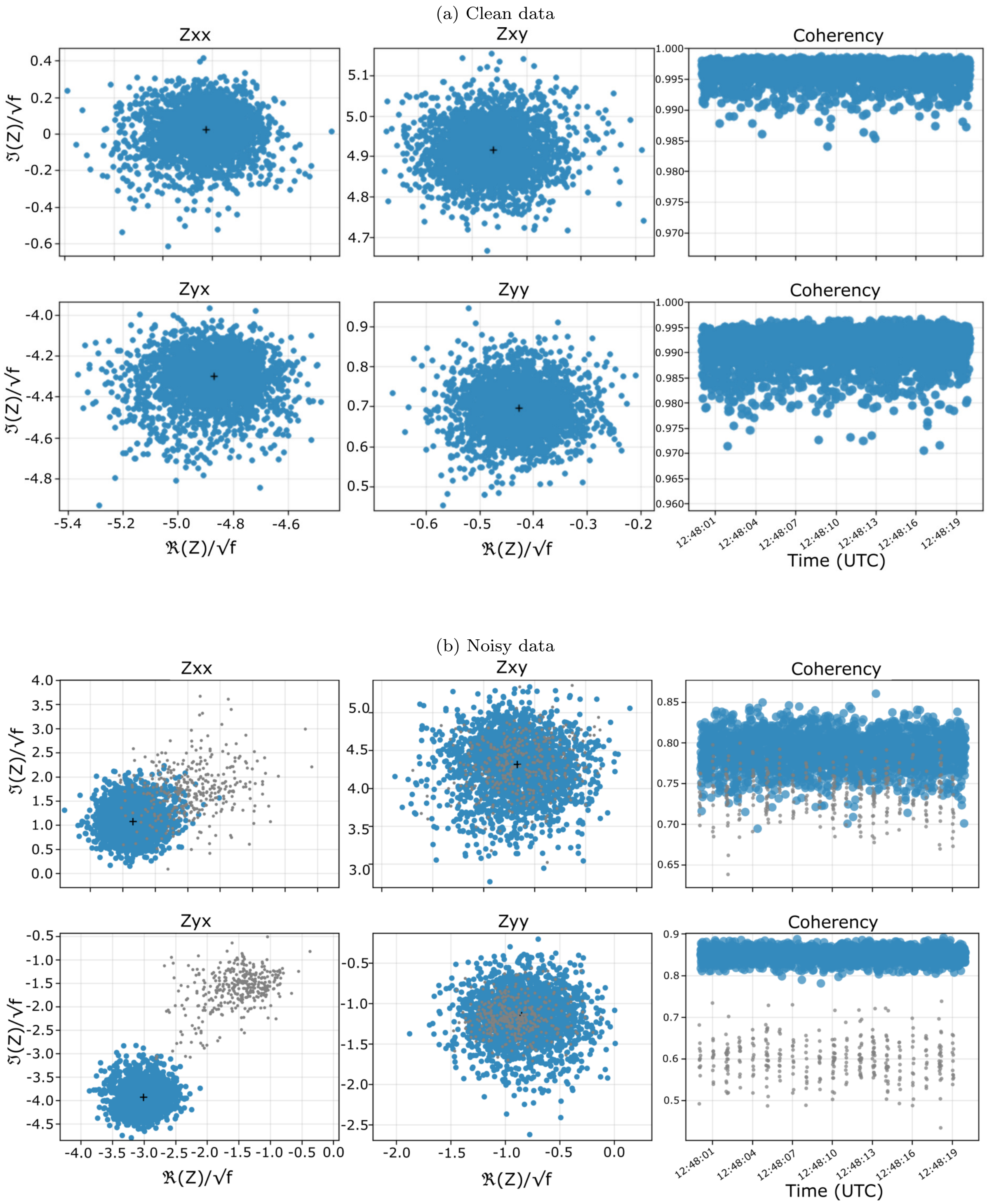


Figure 4.3: Statistical analysis of an example of clean data (a) and noise-affected data (b). Shown are the four impedance tensor components in the complex plane and the corresponding coherency between the magnetic and electric fields.

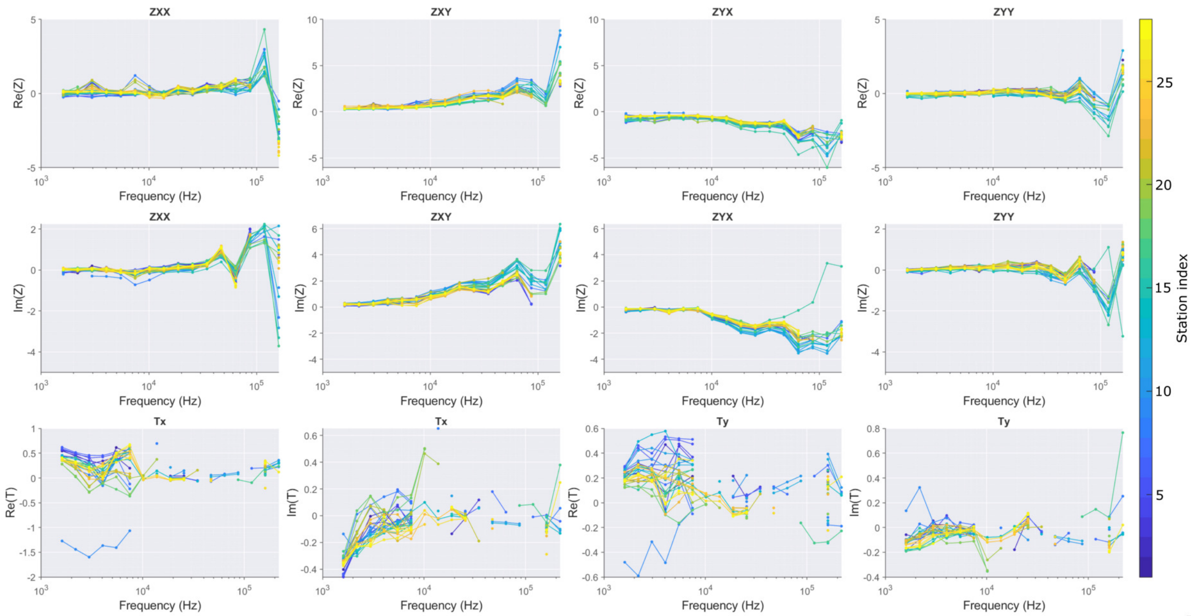
4.1 Data Quality

As already shown in Figure 4.1, several radio transmitters are detected in the recorded data. The spectra of most stations measured during this field campaign are very similar to the example shown in Figure 4.1. However, for the data acquired on Dike 3 with one of the RMT devices, an interfering signal appeared at approximately 100 kHz after about half of the stations had been measured. This signal had a significant impact on the recorded spectra and may have been caused by a problem with one of the cable connections. Although the affected data could still be processed, the increased noise resulted in fewer reliable impedance estimates. Overall, however, the spectra provide sufficient signal quality for estimating the impedance tensor at most stations. A comprehensive assessment of the data quality can only be made after the impedance tensor has been calculated.

Figure 4.4 shows the impedance tensor and tipper function elements for all measured stations at Dikes 1 and 3. For Dike 1 (Figure 4.4a), the data are relatively consistent across most stations. The diagonal components Zxx and Zyy are close to zero, while the off-diagonal components Zxy and Zyx show very similar amplitudes and frequency-dependent behavior but with opposite signs. This similarity, with the components differing mainly in sign, is consistent with the expected response of a predominantly one-dimensional subsurface (Simpson & Bahr, 2005). The tipper data are relatively scattered and difficult to interpret. The data for Dike 3 (Figure 4.4b) show substantially greater variation between stations. This variation may be related to the known lateral heterogeneity of the subsurface, but may also result from the increased noise that affected the transfer functions at approximately half of the measurement stations.

An example of the resulting apparent resistivity and phase curves is shown in Figure 4.5. At first glance, these curves appear rather scattered. Upon closer inspection, however, three distinct frequency ranges with similar characteristics can be identified across all impedance tensor components. At the lowest frequencies, from 1 kHz to 10 kHz, the curves are relatively smooth but exhibit unusual behavior and do not appear to connect continuously with the adjacent frequency range around 10 kHz. Between approximately 10 kHz and 70 kHz, the curves remain smooth for some components, while others display noticeable jumps. At frequencies above 70 kHz, clear discontinuities are present in nearly all curves. This behavior is observed consistently across most measurements, both at different stations within a site and at different survey sites. The inconsistencies at high frequencies are most likely caused by a low-pass filtering effect introduced by the cables used in the Metronix acquisition system (U. Weckmann, personal communication, March

(a) Dike 1



(b) Dike 3

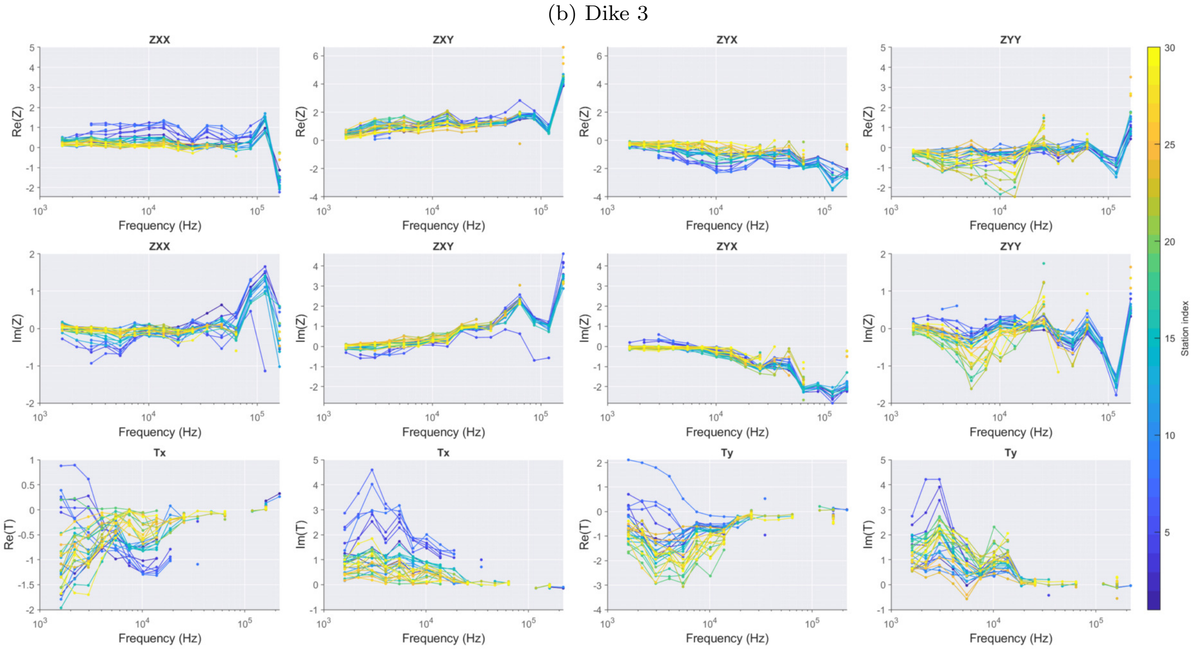


Figure 4.4: Impedance tensor and tipper function elements for all stations of (a) Dike 1 and (b) Dike 3. The color indicates the station index.

2026). At the lowest frequencies (1–10 kHz), the spectral peaks in the amplitude spectra are less pronounced, suggesting that the impedance estimates may be less accurate. There are essentially no suitable radio transmitters in this frequency range, so the source of the observed signal is not entirely clear. Therefore, the results in this frequency range should be interpreted with caution. In addition, the topographic effect of the dike may also influence the measurements. This effect is investigated in more detail in Section 4.2. Another interesting observation can be made by examining the strike directions and induction arrows at different frequencies, as shown in Figure 4.6. For Dike 1 (Figure 4.6a), the induction arrows at 160 kHz point in various directions without a clear pattern. At the lower frequency of 25 kHz, the induction arrows remain small but consistently point away from the dike. The central measurement line is located on the dike crest, whereas the two outer lines are situated at the dike toe. The arrows point away from the dike because the dike represents a relatively conductive structure surrounded by highly resistive air. At the even lower frequency of 2 kHz, the induction arrows become significantly larger and are likely influenced by the adjacent river or the large-scale conductivity structure of the dike and its surroundings. At this frequency, the exact station location becomes less important, and the induction arrows are all oriented in approximately the same direction.

A similar behavior is observed for Dike 3 (Figure 4.6b). In this case, all measurement stations are located on the same side of the dike. At 160 kHz, the induction arrows show no preferred orientation. At 25 kHz, they consistently point away from the dike, while at 2 kHz they become considerably larger, most likely reflecting the influence of the water body located to the northeast of the dike. The strike analysis shows that at low frequencies, the strike direction is closely aligned with the orientation of the dikes. At high frequencies, Dike 3 exhibits no clear preferred strike direction. In contrast, Dike 1 shows a well-defined strike direction that is not aligned with the orientation of the dike. A possible explanation is the presence of an internal conductivity contrast within the dike, which may dominate the local electromagnetic response and give rise to the observed strike direction.

As became apparent from the data, RMT measurements can be strongly affected by noise. This is partly related to the measurement environment, as dikes are often located in areas with substantial infrastructure and anthropogenic electromagnetic noise. However, the quality of the data can also depend on the frequency. At low frequencies, the available signal sources may be unsuitable, for example because they do not sufficiently satisfy the plane-wave assumption. Where the source of the signal cannot be identified,

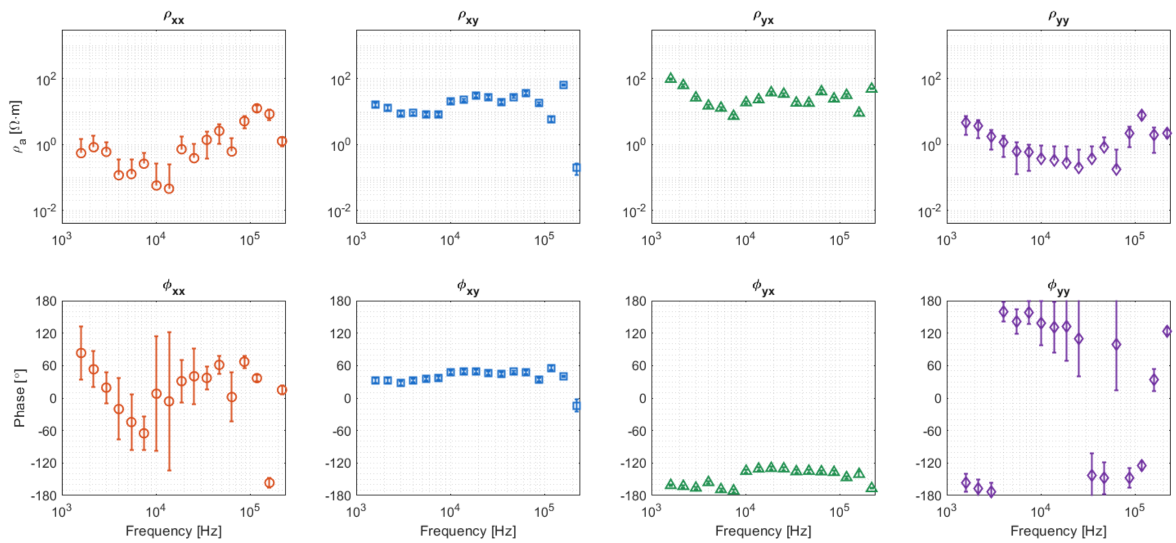


Figure 4.5: Apparent resistivity and phase of all components for a station of Dike 1 (station 1307).

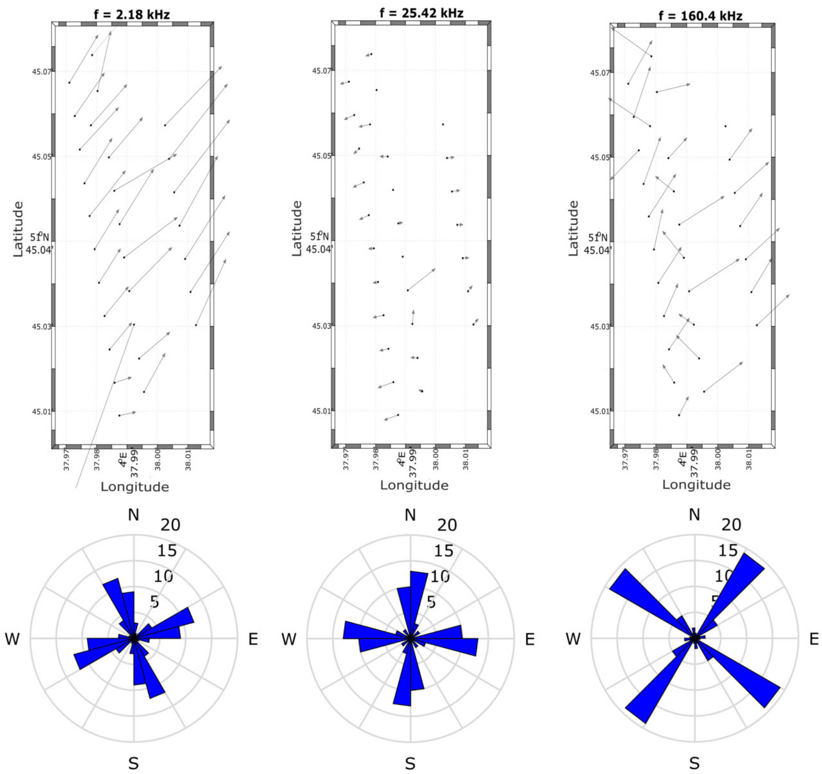
the resulting impedance estimates should therefore be interpreted with caution. At high frequencies, the measurement equipment itself can affect the data by filtering or attenuating parts of the signal. Another potential source of inconsistency is the topography of the measurement area. Topography can affect the electromagnetic field, while uneven terrain may also cause individual electrodes to be differently inclined, meaning that the measured electric field component is not strictly horizontal.

Despite these potential sources of uncertainty, the subsequent data processing is relatively robust when performed carefully, and the impedance estimates are considered reasonably reliable for the available data. The data generally exhibit a predominantly one-dimensional character, which may partly reflect the layered nature of dike structures. However, topographic effects may also contribute to this apparent one-dimensionality, as similar topography-induced apparent layering has been observed in ERT data (Fargier et al., 2014). Nevertheless, this apparent one-dimensionality does not imply that the subsurface can be adequately represented by a 1D model. The absence of a single dominant strike direction and the need to account for topographic effects require a three-dimensional inversion approach.

4.2 3D Effect on RMT Data

To assess the impact of dike geometry on the RMT response, we first identify the frequencies that are expected to be most sensitive to topographic variations. This is done

(a) Dike 1



(b) Dike 3

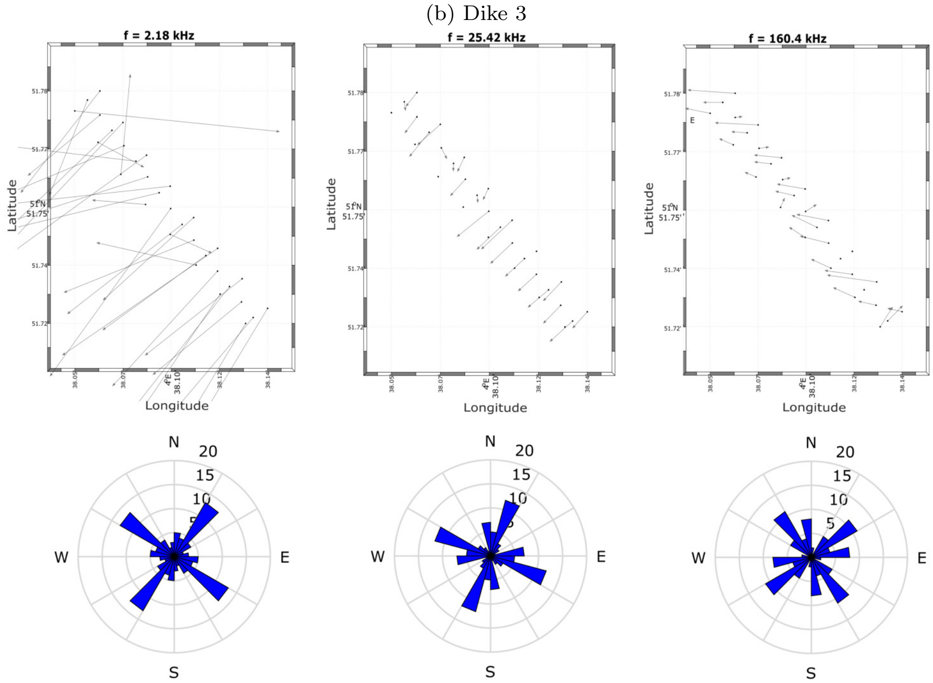


Figure 4.6: Strike analysis and induction arrows for Dikes 1 and 3 at 2 kHz, 25 kHz, and 160 kHz. The induction arrows follow the Parkinson convention and point away from conductive structures. The rose diagrams illustrate the distribution of strike directions across all measurement stations for each frequency. The concentric rings indicate the number of stations associated with each strike direction. These figures were created with mtcode (Darcy et al., 2022).

by estimating the proportion of air, dike material, and sub-dike ground sampled by each frequency, based on its corresponding skin depth. The sensitivity region of each frequency is approximated by a hemispherical volume with a radius equal to the skin depth. While this representation simplifies the actual electromagnetic sensitivity distribution, it provides a useful first approximation of the frequencies for which the dike geometry is expected to have the largest influence on the measured RMT data. In panel (b) of Figure 4.7, the fraction of the sensing volume located in air varies across the entire frequency range. Therefore, no specific frequency can be identified at which the influence of topography becomes negligible. At higher frequencies, the RMT response is dominated by the dike material, as the smaller skin depths result in a shallower investigation depth. Below frequencies of approximately 40 kHz, the sensitivity shifts towards the subsurface beneath the dike, and the measured response is influenced more strongly by the underlying material than by the dike itself.

The influence of the topography on the apparent resistivities of the forward-calculated data for the homogeneous dike model with a resistivity of 28 $\Omega\,$m can be seen in Figure 4.7 (c). Both ρ_{xy} and ρ_{yx} exhibit lower apparent resistivities at lower frequencies.
It should be noted that the analysis of the influence of topography on the data was performed using a ModEM model with a cubical mesh. As a result, the topography could only be approximated, with terrain slopes represented as stepped surfaces rather than smooth gradients. Therefore, the observed topographic effects may not fully reflect those associated with the actual terrain geometry. A more comprehensive investigation of topographic effects would benefit from the use of a finite element code, which enables a more accurate representation of complex topography.

Forward Modeling with River

Next to Dike 1, there is an approximately 250 m wide channel. As this channel is expected to influence the RMT data, it was incorporated into the model. An approximate chloride concentration of 80 mg/L for the channel water was obtained from the Waterinfo database of Rijkswaterstaat (Rijkswaterstaat, 2026). The corresponding electrical resistivity of the water was then calculated to be 5 $\Omega\,$m following the relationship described by Dong, 2013. As no information on the bathymetry of the channel was available, the channel was assumed to have a uniform depth of 10 m for simplicity.

As can be seen in Figure 4.8 (c), the measured data lies within similar ranges as the forward-calculated data. However, the measured data shows considerably more scatter due to measurement noise and the presence of resistivity variations within the actual

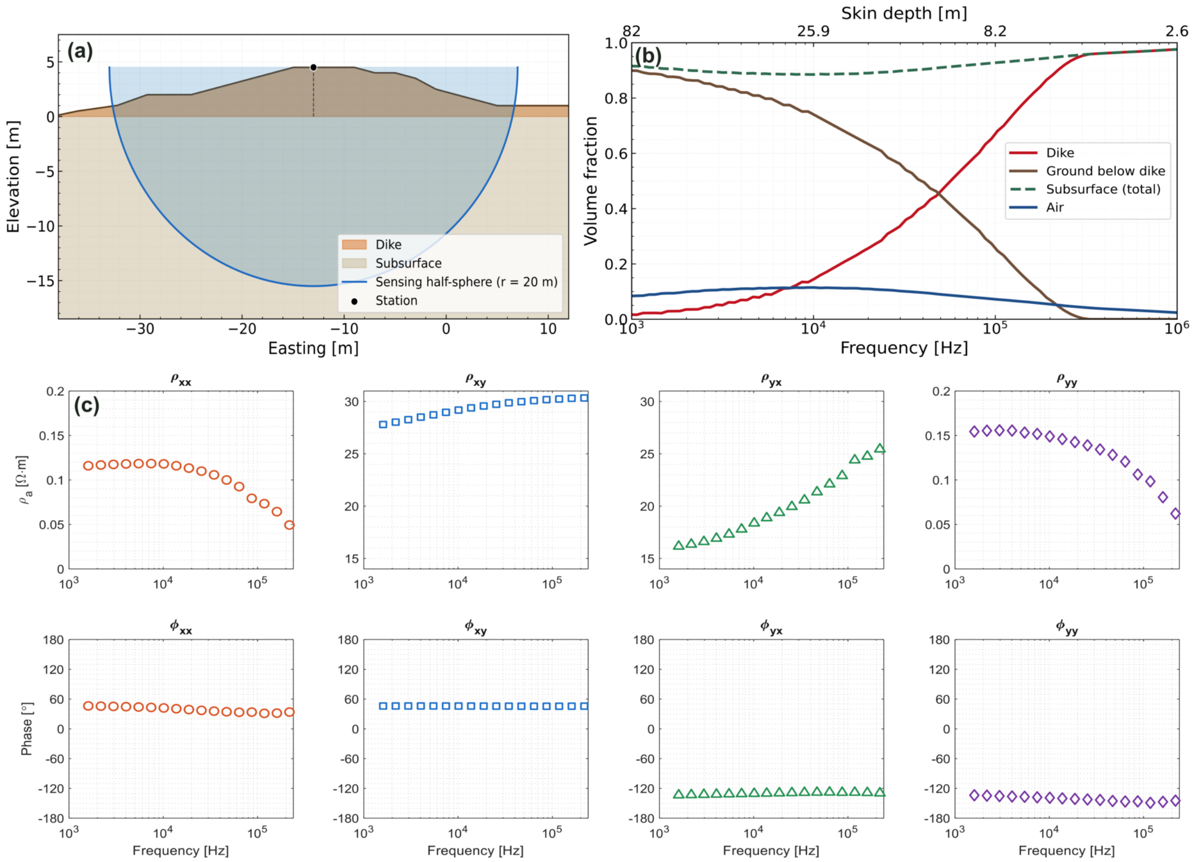


Figure 4.7: Investigation of the geometric effect of dike 1. (a) Schematic representation of the calculation of the sensing hemisphere fractions. (b) Fractions of the sensing volume located in air, dike, and subsurface below the dike as a function of frequency. (c) Forward-calculated RMT response for Dike 1 using a homogeneous 28 Ω m resistivity model with dike topography included.

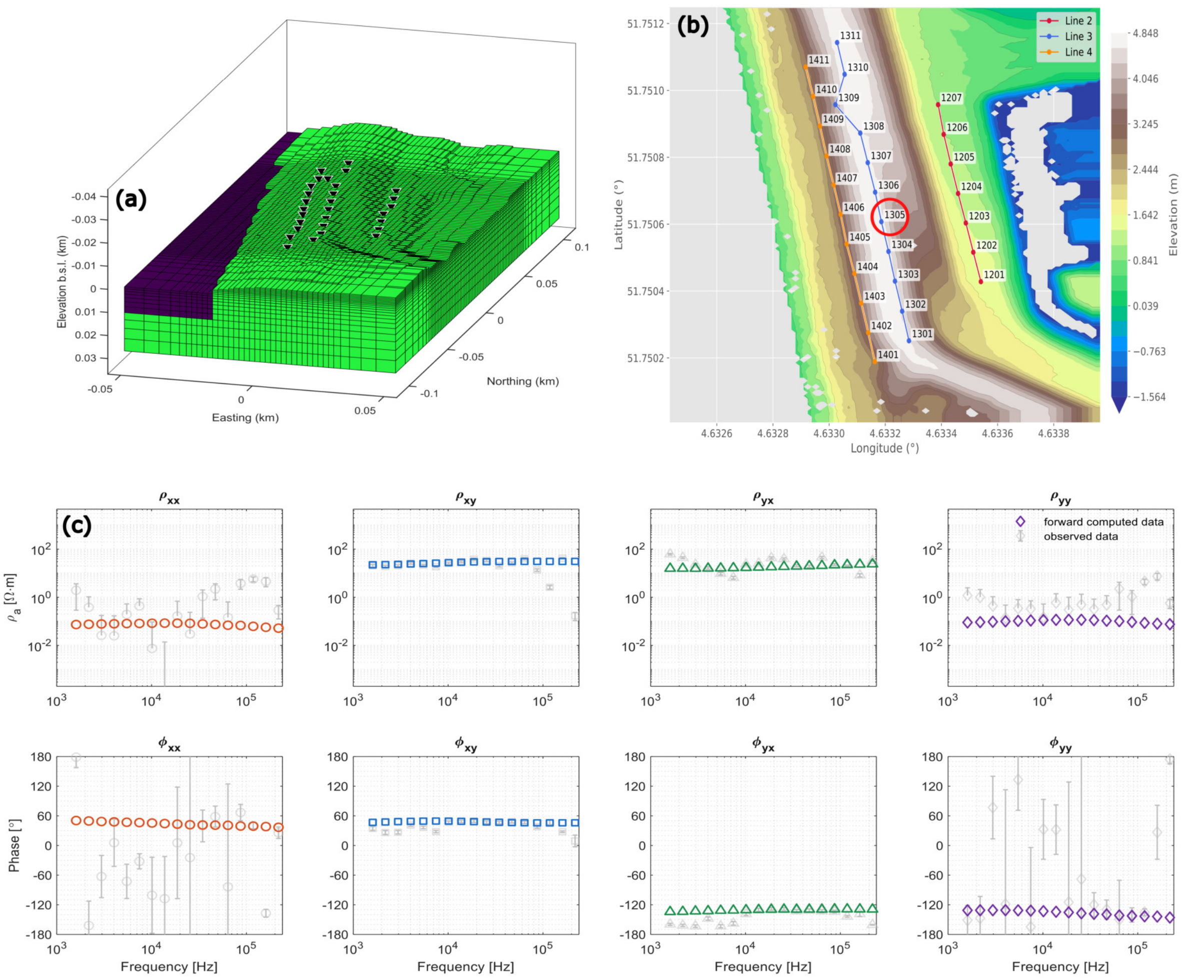


Figure 4.8: (a) Model of Dike 1 including topography, with a homogeneous subsurface resistivity of 28 Ω m and a 10 m deep river assigned a resistivity of 5 Ω m. (b) Topography of the measurement site and the distribution of the RMT stations. No topographic data are shown for areas covered by water. (c) Forward-calculated data from the model shown in (a) at station 1305 (red circle). The gray data points represent the recorded data at the same station.

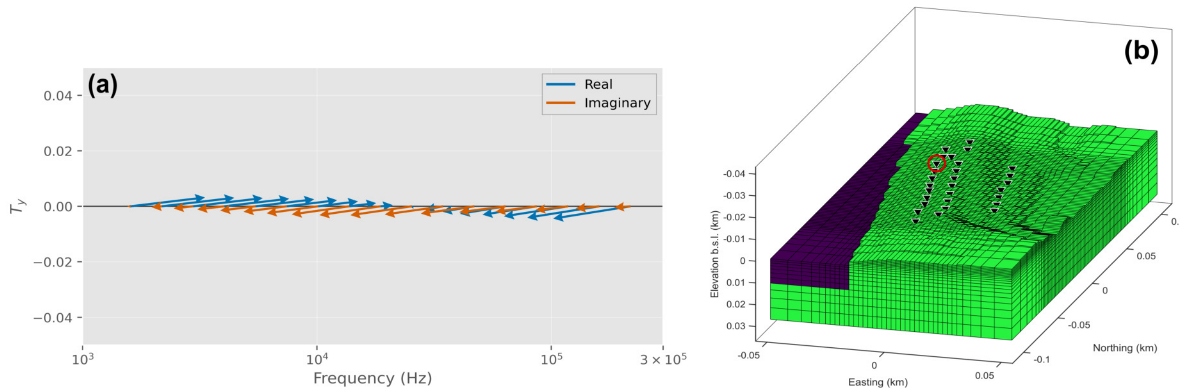


Figure 4.9: (a) Forward-calculated tipper data for station 1406, showing the real and imaginary components. (b) Uniform dike model including the dike topography and the adjacent river. Station 1406 is indicated by the red circle.

dike. Nevertheless, this comparison demonstrates that the relevant physical processes are captured by the forward modeling, as the measured and forward-calculated data show similar characteristics. Therefore, the forward modeling algorithm and the implemented model are considered suitable and can be used with confidence for the subsequent inversion.

The forward-calculated tipper data also show a clear influence of both the dike topography and the adjacent river. Figure 4.9 shows the forward-calculated tipper arrows for station 1406, which is located on the active-side slope of the dike. At high frequencies, the arrows point away from the dike because, relative to the surrounding air, the dike represents the dominant conductive feature. As the frequency decreases, the arrow amplitudes become progressively smaller until, at sufficiently low frequencies, the influence of the river becomes dominant. At these frequencies, the arrows point away from the river, which now represents the most conductive feature in the model. Although the measured data (Fig. 4.6) show the same general behavior as the forward model, with the tipper arrows pointing away from the dike at high frequencies and away from the river at low frequencies, the measured tipper data are considerably noisier and should therefore be interpreted with caution. In particular, the low-frequency arrow amplitudes are much larger than those of the forward model, and the highest-frequency arrows do not exhibit a consistent pattern.

Chapter 5

Inversion

The workflow for the RMT inversions is performed as follows. First, the EDI file of each station is manually edited. This is done by inspecting the phase smoothed D+ solution Beamish and Travassos, 1992 to evaluate whether the apparent resistivity and phase curves are consistent with each other. The D+ solution provides a smooth, physically meaningful fit to the data and is used to assess the consistency between apparent resistivity and phase. Data points where the apparent resistivity and phase show inconsistent behaviour are removed, which is mainly required at the highest frequencies.

Subsequently, the edited EDI files from all stations are combined to create a ModEM (Kelbert et al., 2014) data file. The starting model is generated using mtcode (Darcy et al., 2022), with an initial resistivity of 28 $\Omega\,$m, corresponding to the average resistivity of the ERT model. Tests of the model dimensions showed that a horizontal model extent of 1 km and a depth extent of 500 m are sufficient. The core model area is discretized using cells of 2 m $\times$ 2 m in the horizontal direction and 0.5 m in the vertical direction. Outside the core area containing the measurement stations, the cell sizes are gradually increased by factors of 1.3 horizontally and 1.2 vertically. Finally, the topography is incorporated into the model using elevation data from the Actueel Hoogtebestand Nederland (AHN) dataset (Waterschapshuis, 2022). From this elevation dataset, an xyz topography file is generated and included in the model. The objective function Φ is given as described in Kelbert et al., 2014:

$$\Phi(\mathbf{m},\mathbf{d}) = (\mathbf{d}-\mathbf{f}(\mathbf{m}))^T\mathbf{C}_d^{-1}(\mathbf{d}-\mathbf{f}(\mathbf{m})) + \lambda(\mathbf{m}-\mathbf{m}_0)^T\mathbf{C}_m^{-1}(\mathbf{m}-\mathbf{m}_0) \quad , \tag{5.1}$$

where $\mathbf{d}$ is the observed data, $\mathbf{m}$ is the model, and $\mathbf{f}(\mathbf{m})$ represents the predicted data obtained from the model $\mathbf{m}$. The first term represents the data misfit, where $\mathbf{C}_d$ is the data covariance matrix containing the estimated data uncertainties. The second term represents the model regularization, where $\mathbf{C}_m$ is the model covariance matrix and $\mathbf{m}_0$ is the prior model. The regularization parameter λ controls the trade-off between fitting the observed data and enforcing the constraints imposed by the regularization. It is initially set to 1 and is gradually decreased during the inversion. As λ decreases, the influence of the regularization is reduced, allowing the inversion to place greater emphasis on fitting the observed data. The model covariance matrix defines the preferred model variations and therefore influences the smoothness and structure of the resulting model. For the data covariance matrix, error floors of 10 % for the diagonal impedance elements and 5 % for the off-diagonal impedance elements are used. The prior model $\mathbf{m}_0$ is defined as a homogeneous resistivity model, which is identical to the starting model used for the inversion. While λ describes the relative importance of the regularization compared to the data misfit, the smoothing factors contained within the model covariance matrix control the strength of the regularization in different directions. Values between 0.1 and 1 were tested. The smoothing was kept identical in all directions (x, y, and z) to avoid introducing preferential structures. A smoothing factor of 0.6 was selected as the most suitable value, as lower values resulted in small-scale structures that were not considered physically plausible. The model dimensions were determined through a series of forward modeling tests. Forward responses were computed for models with different spatial extents to identify the point at which further increasing the model size no longer changed the simulated response. At this point, the model boundaries were considered sufficiently far away to have no influence on the results. Based on these tests, model extents of 1 km in the horizontal directions and 500 m in the vertical direction were found to be sufficient. Several inversion tests were performed to evaluate which parts of the dataset provided the most reliable results. Different data configurations were investigated, including the complete dataset, subsets containing only high-, intermediate-, or low-frequency data, manually edited datasets, and different combinations of data components (full impedance tensor, tipper only, and tipper combined with the full impedance tensor). The most plausible inversion results were obtained using manually edited data files, where clear outliers had been removed. Including tipper data in the inversion did not provide significant additional information and had only a minor influence on the resulting models. Furthermore, the tipper data were generally fitted less well than the impedance tensor components.

5.1 ERT Inversion

The ERT data were inverted using SimPEG (Cockett et al., 2015). The inversion employs an inexact Gauss–Newton optimization with weighted least-squares regularization. A homogeneous model with a resistivity equal to the mean apparent resistivity obtained from the measurements was used as the initial model. Since the profiles run along the dike and exhibit no significant topographic variations, topography was not included in the inversion. The ERT inversions were performed using the TreeMesh function implemented in SimPEG. The cells were refined around the electrode locations, resulting in smaller cells near the measurements and progressively larger cells further away. The horizontal and vertical dimensions of the computational domain were selected to sufficiently capture the sensitivity region of the measurements. A relative error of 2% was assigned to all resistivity measurements. For both locations, different starting models and regularization parameters were tested. The resulting resistivity models were highly consistent across the different inversion setups, indicating that the resolved features are robust. The inversions generally converged within 3–10 iterations, reaching RMS values below 2. The inversion results for Dikes 1 and 3 are shown in Figures 5.1 and 5.2, respectively. For both models, the depth of investigation was determined following the approach described by Oldenburg and Li (1999). Two inversions with substantially different starting models but identical regularization parameters were performed. Regions where the two resulting models converged to similar resistivity values were considered to be well constrained by the data. In contrast, regions where the models differed substantially were considered to be poorly constrained, as the inversion remained strongly dependent on the starting model. These poorly constrained regions were masked in Figures 5.1 and 5.2, such that only regions with sufficient sensitivity to the data are displayed. The regularization parameters were typically adjusted after multiple inversion runs until no structural features remained visible in the residual pseudosection, indicating that the main structural information was captured by the inversion model.

The results for Dike 1 (Figure 5.1) show a clearly layered resistivity structure without any pronounced anomalies. A very thin conductive layer is present at the surface, followed by a more resistive layer. At greater depth, the resistivity decreases again, forming a conductive zone, before increasing towards the bottom of the recovered model. Within the conductive layer between approximately 5 and 10 m depth, several variations are visible, indicating distinct zones of higher conductivity. At a horizontal distance of approximately 53 m, the resistive layer appears to be interrupted. A metal fence crosses the ERT line at this location, which is likely responsible for this feature.

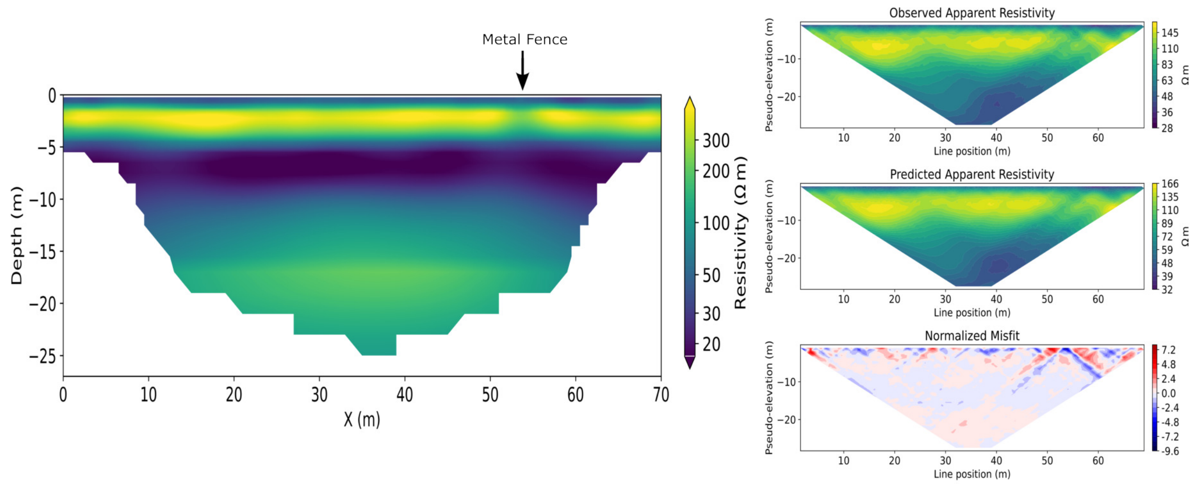


Figure 5.1: Final resistivity model obtained for Dike 1 using the dipole–dipole configuration with 1 m electrode spacing. The inversion converged after four iterations to an RMS of 1.56. The observed and predicted pseudosections, along with their misfit, are shown on the right.

Compared to Dike 1, Dike 3 exhibits substantially greater horizontal variability in the resistivity model (Figure 5.2). The most prominent feature is located between approximately 40 and 80 m along the profile and resembles a graben-like structure. A conductive layer occurs within this region, with its depth varying considerably but generally extending from approximately 5 to 20 m depth. Below this conductive layer, the resistivity increases again, with locally very high resistivity values.

Regarding the ERT inversions, it should be emphasized that although the forward modeling accounts for three-dimensional physics, the resistivity model itself is two-dimensional, and topography was not included. While no significant topographic variations occur along the ERT profiles, the dikes exhibit pronounced topographic variations perpendicular to the profiles. These variations are therefore not represented in the model and may influence the inversion results. This effect should not be underestimated (Fargier et al., 2014).

5.2 RMT Results Dike 1

For Dike 1, most inversions converged to RMS values between 2.5 and 5. The inversions typically required around 20 to 70 iterations. The inversion results were strongly influenced by the choice of the topography file. Several approaches were tested, including using the actual topography for the entire model domain, applying topography only to the dike while keeping the surrounding areas flat, and using the dike topography combined with an interpolation of the missing areas to avoid abrupt changes in elevation.

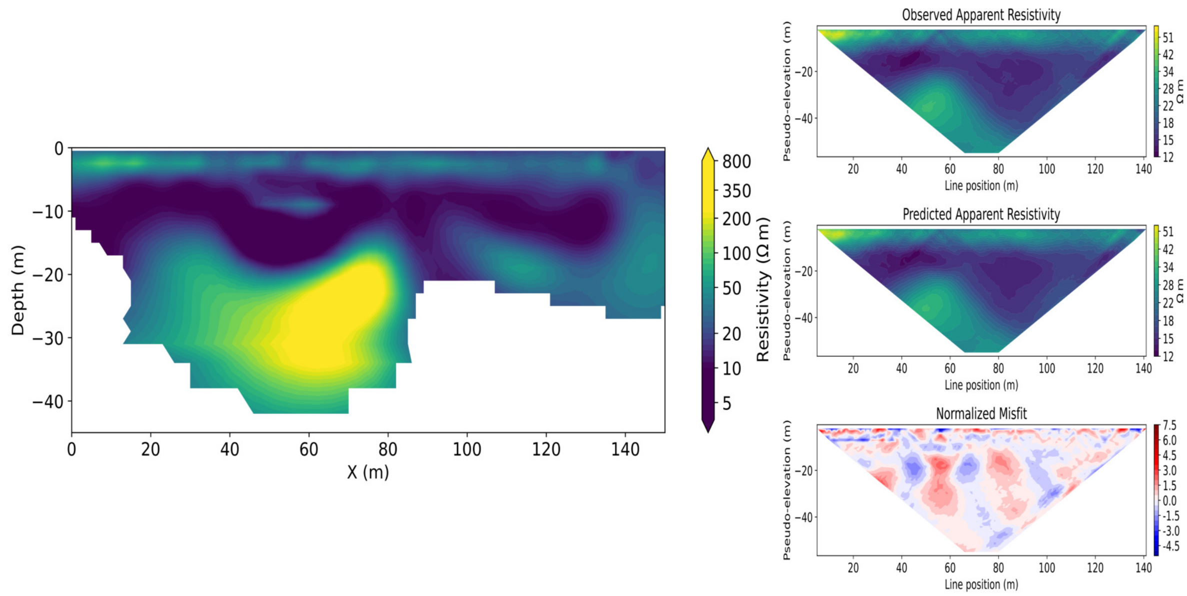


Figure 5.2: Final resistivity model obtained for Dike 3 using the dipole–dipole configuration with 2 m electrode spacing. The inversion converged after four iterations to an RMS of 1.73. The observed and predicted pseudosections, along with their misfit, are shown on the right.

It was found that large artificial steps in the topography resulted in poorer inversion results; therefore, simply restricting the topography to the dike area did not provide satisfactory results.

The software package mtcode (Darcy et al., 2022) automatically assigns ocean resistivity to all areas where the topography is below 0 m, by adding an ocean layer between the surface elevation and sea level. However, since large parts of the Netherlands have elevations below sea level without representing ocean areas, this behavior had to be modified in the code. The computational grid for Dike 1 consisted of $96 \times 64 \times 40$ cells in the x, y, and z directions, respectively. The final inversion model is shown in Figure 5.3. The inversion converged with a mean RMS value of 3.83 after 21 iterations, with the data fit of an example station shown in Figure 5.4. The resulting resistivity model exhibits a conductive–resistive–conductive–resistive layering. Inversions performed with different settings or different data files resulted in slightly varying models. However, all inversion results consistently show clear conductive anomalies at approximately 15 m depth, located at 20 m and 45 m along the dike crest. The horizontal slice shown in Figure 5.3 indicates that these conductive features form elongated structures oriented approximately perpendicular to the dike. It is likely that the true extent of these features is smaller than represented in the inversion model. However, they indicate the presence of possible conductive pathways crossing the dike.

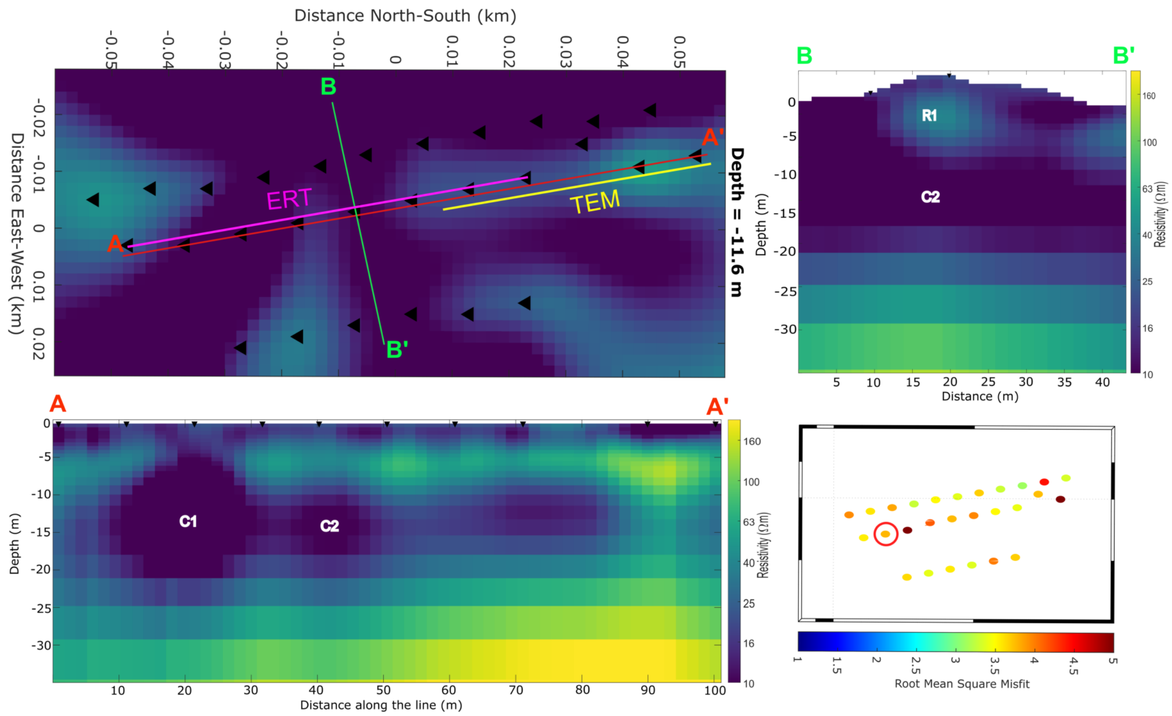


Figure 5.3: 3D resistivity model from the RMT inversion, including a horizontal slice at 11.6 m depth, two vertical cross-sections, and a map with the RMS misfit for each station. The labels C1 and C2 indicate the two distinct conductive anomalies.

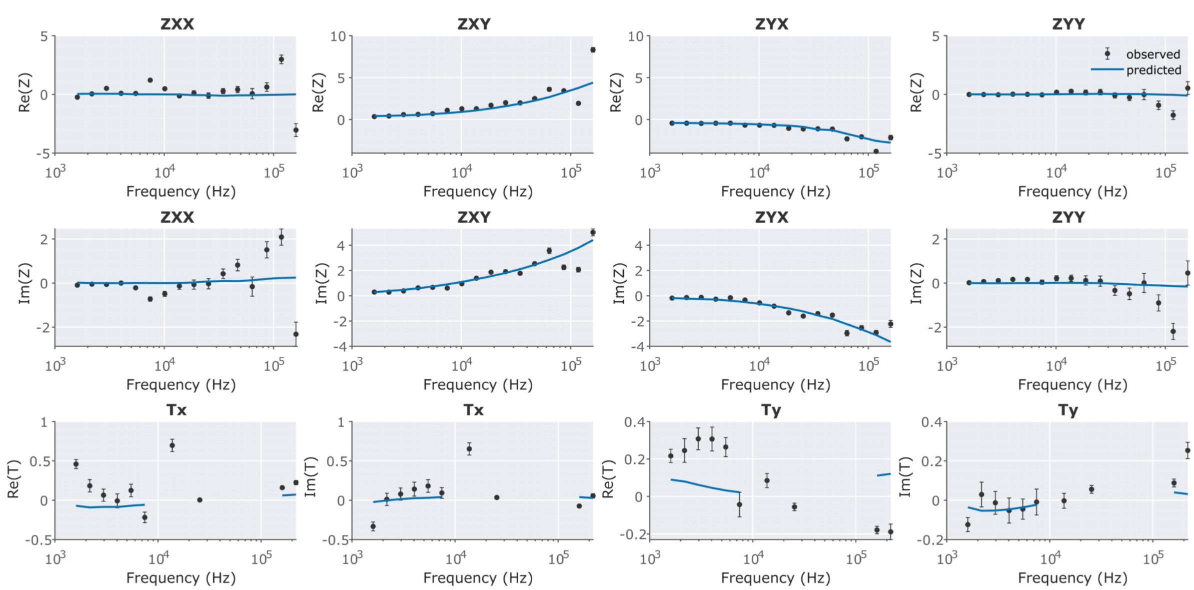


Figure 5.4: Data fit for station 1302 for the real and imaginary components of all impedance tensor elements and the tipper.

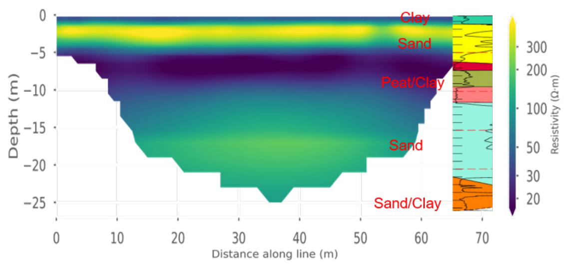


Figure 5.5: Resistivity model obtained from a 2D ERT inversion along the dike crest. The interpreted lithological profile is based on a CPT and was provided by Tom de Gast (Waterschap Hollandse Delta).

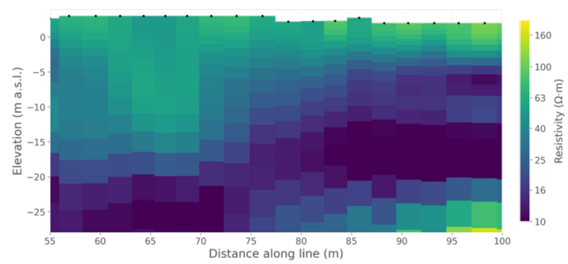


Figure 5.6: Profile of 1D TEM inversion results along the dike crest. The data were acquired and inverted by Jörn Löhken (geofact) using an sTEM system.

As shown in Figure 5.4, the predicted impedance tensor elements reproduce the general trends of the observed data, while the small-scale variations in the observations are not reproduced. The highest frequencies are not fitted accurately, which is expected given the high noise levels at these frequencies, where the observed data points show substantial scatter. The tipper components are fitted poorly and therefore provide limited additional information for the inversion. Excluding the tipper data from the inversion resulted in very similar models, further indicating their limited influence on the inversion results.

5.3 Interpretation Dike 1

Figures 5.5 and 5.6 show the electrical conductivity profiles along the dike crest obtained from ERT and TEM data. The TEM data is inverted in 1D and the crossection in Figure 5.6 is a sequence of these 1D inversions. Looking at Figure 5.3, the RMT inversion shows a resistivity layering, with alternating conductive and resistive layers. From the surface, the model shows a conductive layer followed by a more resistive layer, then another conductive layer and a deeper resistive layer. Based on the CPT interpretation in Figure 5.5, these layers correspond to a clay cover, a sand layer, a clay–peat mixture, and a deeper sand layer, respectively. The ERT inversion reveals a similar layered structure, which agrees well with the interpretation of the CPT data. The RMT inversion reveals two distinct conductive anomalies. However, their interpretation should be treated with caution, particularly regarding their depth and extent. The ERT result also shows a conductive feature at approximately the same horizontal position, but at a shallower depth. While the most conductive features in the ERT model occur at approximately 5–10 m depth, the corresponding conductive features in the RMT model are located deeper, at approximately 10–20 m depth. The TEM data provide limited information in the first part of the profile, between 55 and 75 m along the RMT line, where the measurements are strongly affected by a metal fence. Elsewhere, however, the TEM model shows a clear conductive feature at approximately 15–20 m depth, which is more consistent with the RMT model than with the ERT model. The difference in the depth of the conductive features between the ERT and RMT/TEM models could be related to the different physical principles of the methods, with ERT measuring a potential field and RMT/TEM relying on electromagnetic induction. This is discussed in more detail in Section 6. In the ERT model, the conductive feature appears to be confined to the clay–peat layer, whereas in the RMT model, the anomaly extends across several layers and appears considerably larger. RMT is highly sensitive to conductive features. However, due to the regularization applied during inversion and limitations in data quality, the exact geometry of these features cannot be reliably resolved. As multiple inversions were performed using different inversion settings, topography files, and data subsets, the features C1 and C2 can be considered robust. Their horizontal positions and the orientations of the high-conductivity zones remained similar across all inversions. However, their depth, size, and absolute resistivity values varied. The horizontal slice at 11 m depth shows that the anomalies extend approximately perpendicular to the dike, connecting the active and passive sides. However, the high uncertainty in depth makes the interpretation of this feature difficult. Seepage structures would be expected to occur at approximately the depth of the river water table or slightly below, which is consistent with the depth indicated by the ERT. If the conductive feature is instead located deeper, as suggested by the RMT, it could represent natural variations in the stratigraphy below the dike rather than a dike-related feature or instability. Notably, the orientation of the conductive feature C1 closely matches the strike direction recovered at high frequencies in Figure 4.6a. This suggests that the conductive feature is sufficiently strong to locally influence the estimated strike direction.

5.4 RMT Results Dike 3

As the tests of the grid sizes and model dimensions were already performed for the first dike and it was assumed that we are in a similar setting, the same cell sizes and number of padding cells around the study site were applied as for Dike 1. As the lines were a lot longer than for Dike 1, this resulted in more grid cells, namely $100 \times 96 \times 40$ (384'000 cells compared to 245'760 for Dike 1). The RMT inversion results are shown in Figure 5.7. The inversion took 30 iterations and ended with an RMS of 7.67. This high RMS can be attributed primarily to the tipper data, which could again not be fitted accurately (Figure 5.8). Furthermore, approximately half of the stations at Dike 3, comprising half

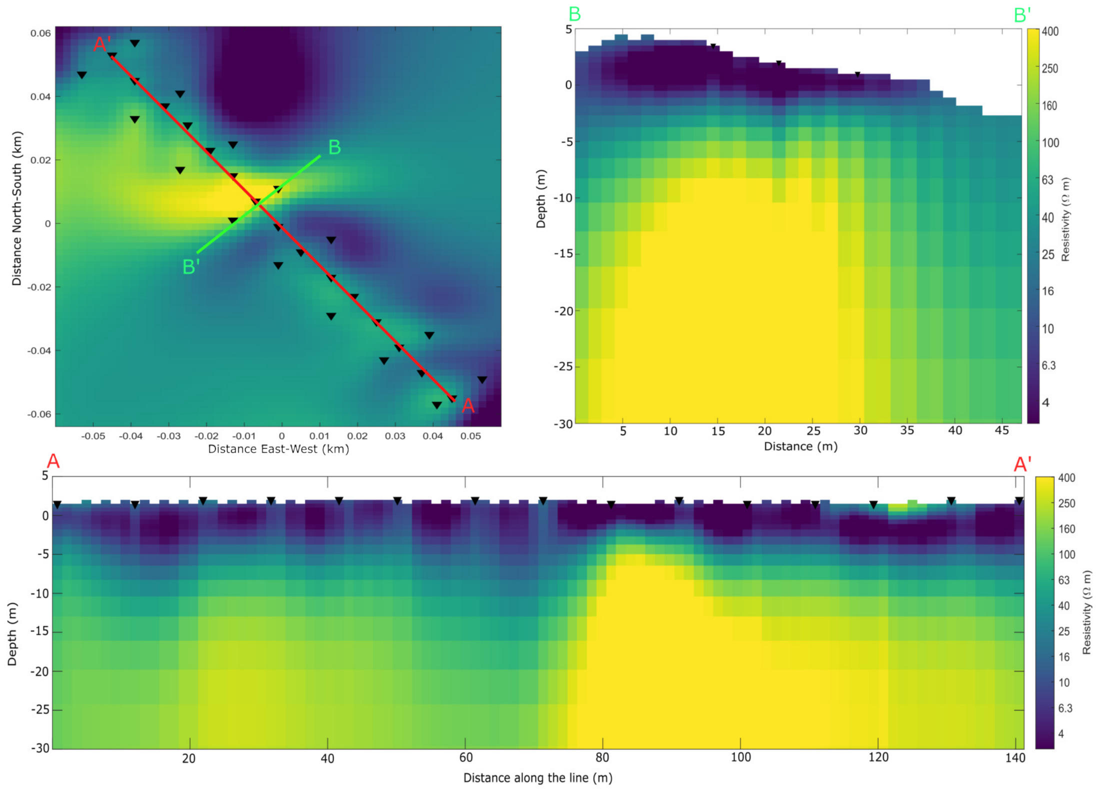


Figure 5.7: 3D RMT inversion of Dike 3. Displayed is a horizontal slice at 7 m depth, a longitudinal profile along the passive side of the dike and a cross section perpendicular to the dike.

of Line 2 and the entirety of Line 3, were severely affected by noise. This likely affected the inversion's ability to fit the data. At the stations with better data quality, the impedance tensor elements were fitted reasonably well. In Figure 5.9, the highest misfit values are observed at the stations towards the northwest. This should be kept in mind when interpreting the resistive area in the northern part of the horizontal slice shown in Figure 5.7. In addition, a very shallow resistive feature is visible at approximately 120–130 m along the profile. While this feature may represent a genuine subsurface variation, it could also be an artifact resulting from the high noise levels in the high-frequency data.

Overall, two distinct zones can be identified. A conductive layer overlies a resistive area, with limited internal structure resolved within either zone. However, the depth of the transition between the conductive and resistive zones varies considerably along the dike profile. The resistive area near the surface at around 120 m is not supported by any other dataset and may be an artifact caused by high-frequency noise.

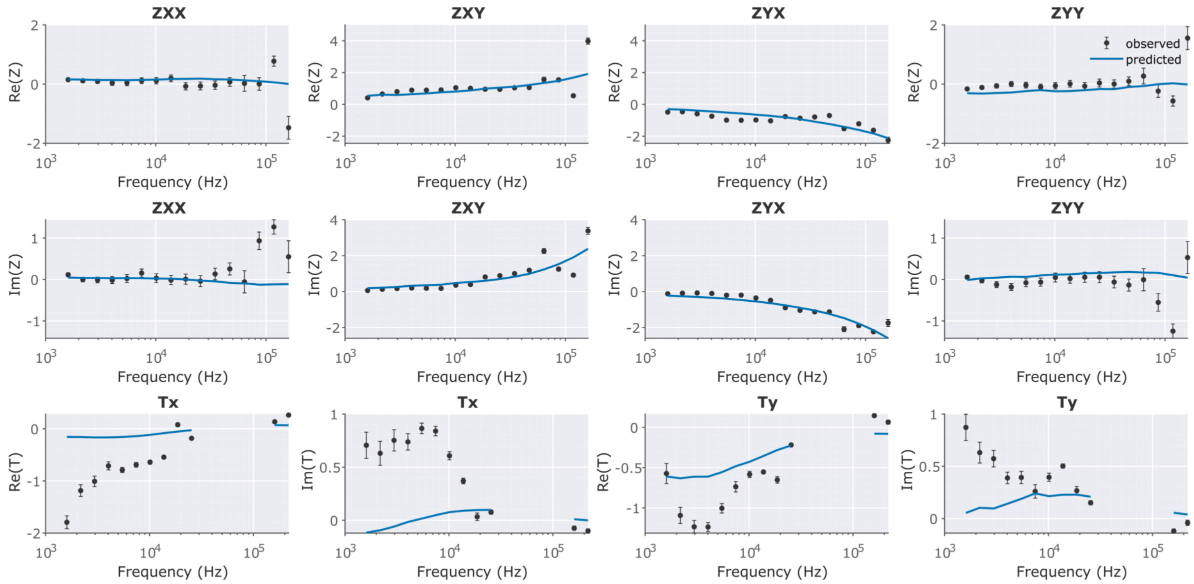


Figure 5.8: Data fit for station 3201, showing the real and imaginary parts of the impedance tensor estimates and the tipper components.

5.5 Interpretation Dike 3

First, it should be noted that the interpretation of Dike 3 is associated with considerable uncertainty, as an RMS of 7.67 indicates a relatively poor fit to the data. Nevertheless, the ERT data and prior knowledge of the internal structure provide additional information that can be used to assess which features are visible in the RMT model. One clear difference between Dike 3 and Dike 1 is that Dike 3 appears to consist predominantly of conductive material. In contrast to Dike 1, where the sand core of the dike could be clearly identified, there is little variation in resistivity within Dike 3. This is consistent with the subsurface model shown in Figure 3.4, which indicates that the dike mainly consists of clay. The only more resistive materials are the sand layer added on the passive side and the sand layer at depth. While we do not see much variation in the upper parts of the model, the contact between the upper clay layers and the lower sand layer is well visible in the RMT inversion. For the ERT inversion in Figure 5.2 we do see much more features in the top 20 meters. It is clearly visible that the uppermost five meters are more resistive, and that there is a very conductive layer below. The resistive area on top is probably the added sand, and the conductive layer below consists of peat and clay as suggested in Figure 3.4. The graben structure between approximately 40 and 80 m along the profile is also clearly visible. Furthermore, both the ERT and RMT models show that the depth of the clay layer increases towards the northwestern part of the dike. The TEM model in Figure 5.10 shows that the added sand layer is well resolved

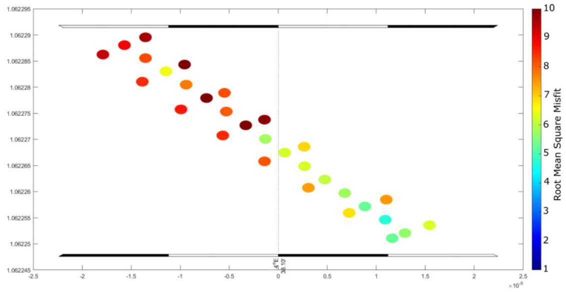


Figure 5.9: Misfit map of the RMT inversion of Dike 3. The color shows the misfit of the individual stations. The average final misfit is 7.67.

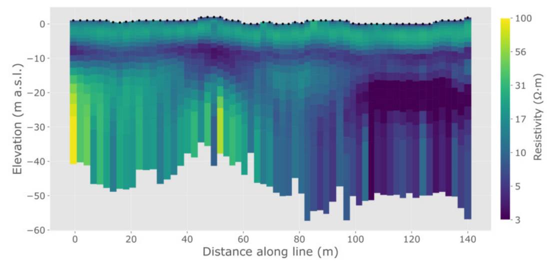


Figure 5.10: Profile of 1D TEM inversions along line 2 of Dike 3. The data were acquired and inverted by Jörn Löhken (geofact) using an sTEM system.

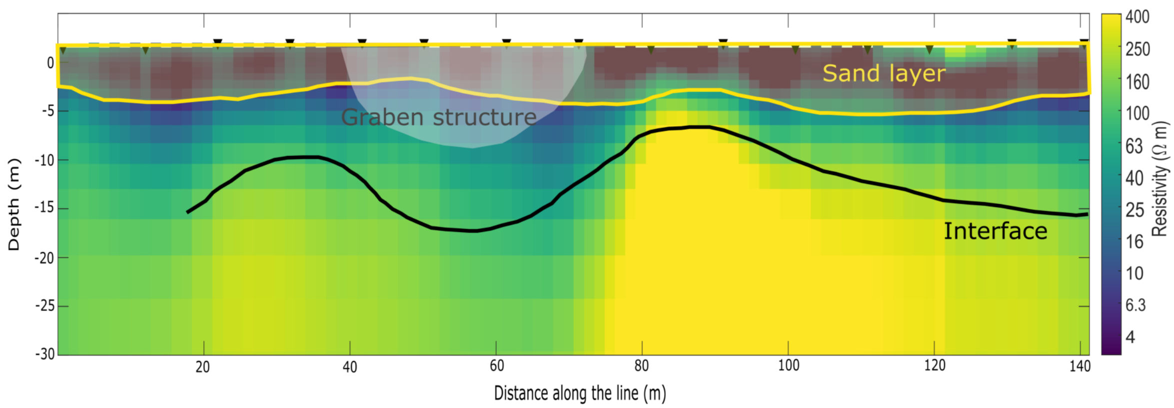


Figure 5.11: RMT inversion model with structures identified in the ERT and TEM models overlaid. The interface and graben structure are interpreted from the ERT model, while the sand layer is identified from the TEM model.

by the TEM data. However, below this layer, the model provides limited information, and the graben structure is not clearly visible. Figure 5.11 shows the RMT model, with features identified in the ERT and TEM models overlaid for comparison. The interface between the conductive clay layers and the underlying sand layer is well matched between the RMT and ERT models. However, the sand layer and the graben structure are not resolved in the RMT model.

Chapter 6

Discussion

There are several aspects that need to be considered before a final conclusion can be reached on whether RMT is a suitable method for dike monitoring. Most features identified in the RMT inversion could, to some degree, be validated by the ERT results. Furthermore, the high noise level did not result in widespread artifacts in the RMT inversion, but rather limited the amount of reliable information contained in the affected data. The conductive zones detected in Dike 1 were, however, most likely overestimated in both their lateral extent and thickness. This can be attributed to the diffusive nature of RMT and the applied regularization, which tend to smooth strong resistivity contrasts and spread conductive features beyond their actual boundaries. However, RMT provided clear indications of the 3D orientation of the conductive feature for Dike 1, which is consistent with the expected orientation of seepage pathways or piping features. This suggests that, if further investigations of this dike are conducted, they should focus on the areas where these conductive features are located.

It has been demonstrated that the topography of the dike has a significant impact on the RMT data. Although the dike topography was represented as accurately as possible in the forward model, the cubic cells used in ModEM limited the representation of the sloping sides of the embankment. The geometry of the dike affects the propagation and distribution of the electromagnetic fields within the subsurface, particularly where the field interacts with the sloping boundaries of the embankment. Consequently, even small differences between the actual and modeled topography can affect the resulting RMT response and the position of structures recovered by the inversion. This effect could potentially contribute to the mismatch in the depths of the features identified in the ERT and RMT models.

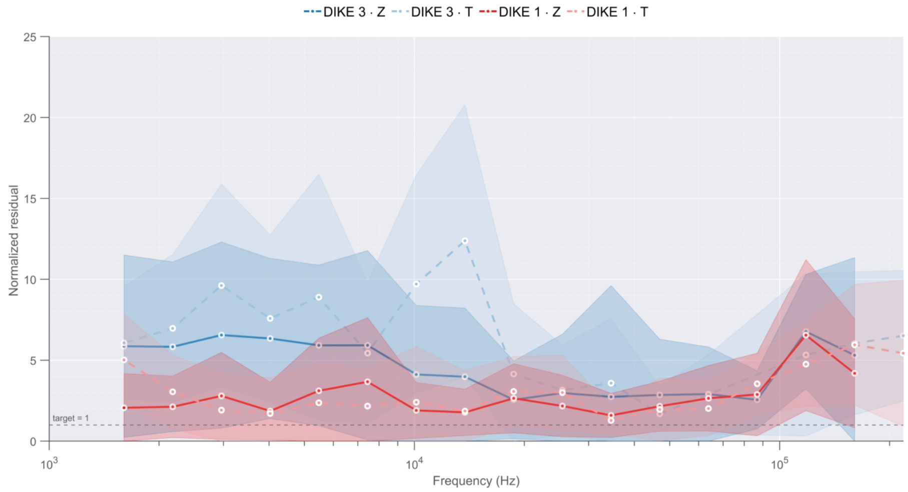


Figure 6.1: Comparison of the residuals at different frequencies. The four lines show the mean residuals of the impedance tensor and tipper elements for Dikes 1 and 3. The shaded areas indicate one standard deviation around the mean residual at each frequency, calculated from the residuals of all stations.

Figure 6.1 shows the residuals between the observed and predicted data for each frequency. The residual is calculated as the difference between the observed and predicted data, normalized by the corresponding data error. Thus, a residual of 1 indicates that the difference between the observed and predicted data is equal to the estimated data error. Overall, the best fit is achieved at frequencies between 10 kHz and 100 kHz. This is consistent with the observations from the measured data, where the higher frequencies show substantial scatter, while the lower frequencies exhibit behavior that is not necessarily physically meaningful. For Dike 1, the low-frequency data are nevertheless fitted relatively accurate. The strong noise at the higher frequencies is considered to be the main issue affecting the inversion models. For electrical resistivities expected within the dikes, the skin depth at the highest frequency that was measured is already around 5 m, which is approximately the depth of the dike itself. This means that the highest frequencies are particularly important for obtaining information about the interior of the dike. Resolving this issue should be one of the main priorities for future work. One possible approach would be to investigate whether cables with less attenuation at the highest frequencies are available. Alternatively, a different instrument with a higher sampling rate could be used to record higher frequencies. Another approach to obtain more information about the shallow parts of the dike would be a joint inversion of the RMT and ERT data, as ERT is able to resolve shallow structures well.

As demonstrated by the synthetic modeling, RMT is more sensitive to conductive features than to resistive ones. This is also reflected in the field data. The conductive feature in Dike 1 was clearly visible in the RMT model, whereas the added sand layer in Dike 3 was not resolved. However, the poor resolution of the sand layer cannot be attributed solely to its resistive nature. The layer is located very close to the surface, where the high-frequency data are required to obtain sufficient resolution. The poor quality of these data therefore likely also contributed to the inability to resolve the sand layer. Regarding the detectability of resistive and conductive features, it should be noted that this depends strongly on the materials from which the dike is constructed. In a dike with a sandy core, for example, infiltrating water could produce a sufficiently strong resistivity contrast to be detected by RMT. In contrast, Dike 3 is predominantly composed of conductive clay. In such cases, low-salinity water, as commonly found in many parts of the Netherlands, can have electrical resistivity values similar to those of the surrounding dike material. Consequently, the resistivity contrast between the infiltrating water and the surrounding dike may be too small to allow the feature to be distinguished based on RMT data alone. The detectability of instability features also depends on the hydrological state of the dike, as water-filled voids produce a stronger resistivity contrast than dry, air-filled voids.

RMT is a relatively elaborate method, and both data acquisition and processing require careful attention to several methodological aspects. Data acquisition needs to be performed with great care, as the electrodes must be positioned with a consistent orientation. In addition, infrastructure and topography can complicate electrode placement and make it difficult to maintain the required measurement geometry. Great care is also required when processing RMT data. In particular, the selection of frequency bins has a strong influence on the resulting impedance tensor estimates. The bins need to be sufficiently wide to include enough signal from different directions, while being narrow enough to retain the frequency-dependent information. Furthermore, the impedance estimates and their distributions should be carefully inspected before selecting the appropriate statistical measures for estimating the impedance tensor.

The tipper data acquired in this project could not be interpreted to a satisfactory degree. The expected effects of the dike topography and surrounding water bodies were visible in the tipper data, but no additional information about the dike interior could be obtained. The high-frequency tipper data were too noisy to provide reliable information about the subsurface. Nevertheless, this does not mean that no information can be gained from tipper data. The measurement setup used in this project was not specifically optimized for the interpretation of the tipper. While the electric field was measured at 10 m intervals, the magnetometer was only relocated every 20 m, based on the assumption that the magnetic field varies more gradually than the electric field. For airborne measurements, the magnetic field could be sampled at much smaller spatial intervals, which could potentially provide more detailed information from the tipper data.

Overall, when comparing ERT, TEM, and RMT at the two investigated sites, ERT provided the most information on the subsurface structure. TEM showed limited sensitivity at greater depths and, at Dike 1, was strongly affected by an electrical fence. Such sources of electromagnetic noise are common in urban environments and can make TEM surveys of dikes challenging. A major advantage of the RMT data is that they were evaluated using a 3D inversion, which allowed the spatial extent and geometry of subsurface features to be investigated. A 3D approach is particularly useful for dike investigations, as it allows the complex topography and three-dimensional structure of the dike and the internal structures to be taken into account.

The acquisition time of RMT and ERT measurements is comparable, but RMT requires considerably more processing. Therefore, despite the similar field effort, a dense 3D ERT survey may still be a more practical approach for detailed dike investigations. The main potential of RMT lies instead in its high sensitivity to conductive structures. Combining RMT with ERT could therefore provide complementary information and improve the sensitivity to conductive anomalies associated with features such as piping or seepage.

Chapter 7

Conclusion

The measurements in this study showed that electrical resistivity varies significantly within embankment structures depending on their material composition and water saturation. These variations can provide indications of internal heterogeneity and potential instability features. The results also showed that RMT can provide useful information about the subsurface even in challenging urban environments, where data acquisition can be difficult and significant noise can affect the measurements. The inversion of the RMT data with ModEM was able to fit the data reasonably well and produced physically meaningful results. The 3D inversion provided information on the spatial orientation of conductive features that would be difficult to obtain from a single 2D profile. However, the resolution in the shallow subsurface was limited, mainly due to the poor quality of the high-frequency data. This limitation made it difficult to resolve small-scale features within the main body of the dike. In addition, dike topography was found to have a significant influence on the inversion results. Future studies should therefore address both limitations by improving the high-frequency response of the measurement system and investigating alternative inversion codes, ideally using a finite-element approach that can represent the actual dike topography more accurately.

To conclude, RMT is able to provide information about the subsurface resistivity structure of embankments. Particularly in dikes containing a sand or gravel core, seepage pathways may produce a sufficient resistivity contrast to be detected by RMT. However, the effort required relative to the information gained needs to be considered. If the current limitations related to the high-frequency transfer functions and the representation of topography can be addressed, I consider RMT a useful complementary method alongside ERT and/or geotechnical measurements for detailed investigations. Nevertheless, due to the time-intensive processing required for RMT, its effectiveness for detailed dike investigations may remain lower than that of ERT. With a comparable overall effort, a dense ERT grid could potentially be acquired and inverted, providing a model with substantially higher spatial resolution. The main value of RMT therefore lies not in replacing ERT, but in providing complementary information, particularly on conductive structures that may be associated with seepage or piping. To assess the potential of RMT as a drone-based method, additional tests should be conducted using survey layouts that more closely represent the type of data that could be acquired with a drone.

Acknowledgments

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Data Availability

All code used in this thesis is publicly available in the accompanying GitHub repository: https://github.com/ramonvanoli/RMT_dike_investigation. The data used in this study will also be made publicly available in the near future.

Use of AI Tools

Parts of the text in this thesis were edited and refined with the assistance of the AI language model ChatGPT (OpenAI) to improve clarity and academic phrasing. Scripts for model and data plotting and editing were developed with AI support using Auto in Cursor and Claude Sonnet 5 (Anthropic). All scientific content, analyses, and interpretations were produced solely by the author.

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Appendix A

Additional Dike Locations

A.1 Beaver - Dike 2

The second survey site was located in an area where beaver activity had been observed. A local wildlife manager identified the entrance of a beaver burrow. The survey design was adjusted accordingly, with the ERT profiles positioned such that the inferred burrow entrance was located near the centre of the lines.

Preliminary field inspection of the ERT pseudosections indicated that the burrow was not clearly resolved in the data. Given that ERT generally provides higher sensitivity to resistive structures than RMT, the investigation at this site was not prioritized further, and only three RMT stations were acquired along Line 1. In addition, a second ERT profile (Line 2) was measured along the crest of the dike. The survey layout is shown in Figure A.1. This layout was chosen to test whether incorporating RMT station data

Table A.1: Overview of acquisition parameters for Dike 2.

Parameter	**Value**
Location	51°42’52.69”N, 4°34’55.40”E
Date	10 March 2026
Number of measurements	3 RMT stations, 2 ERT Lines
ERT Electrode spacing	1 m
RMT Electrode spacing	10 m
RMT sampling frequency	524,288 Hz
Weather conditions	slight rain
Instruments	2 × ADU-08e (Metronix), 1 × ERT system (IRIS Instruments)
Measurement sequence	ERT Line 1, ERT Line 2, RMT Line 1

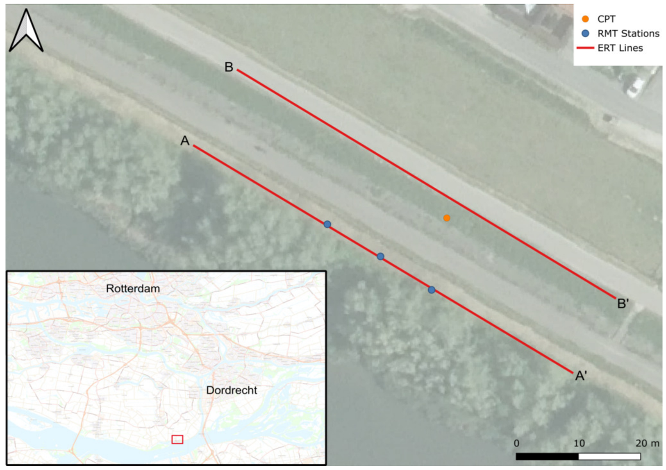

Figure A.1: Survey layout of Dike 2, including RMT station positions and ERT profiles.

into the inversion can improve the ERT model. Joint inversion could not be carried out in the scope of this Masters thesis and would be a possible next step of the project. The resulting resistivity profiles for Lines 1 and 2 are displayed in Figures A.2 and A.3, respectively. Figure A.4 shows the RMT data obtained at Dike 2

A.2 Leidschendam - Dike 4

The last location was selected because ERT data from this site were already available. The dike has been the subject of numerous previous investigations, and the ERT profiles revealed a distinct electrical resistivity anomaly within the structure, making it an interesting target for further study. The site is located in an urban environment near a railway station. Consequently, one of the main objectives was to assess whether RMT can provide reliable results in areas characterized by elevated levels of anthropogenic electromagnetic noise. Because the dike is relatively narrow, an electrode spacing of 10 m was not feasible along the entire profile. Therefore, the electrode spacing was reduced locally to approximately 5 m where necessary. The survey layout is shown in Figure A.5, and the RMT data are shown in Figure A.6.

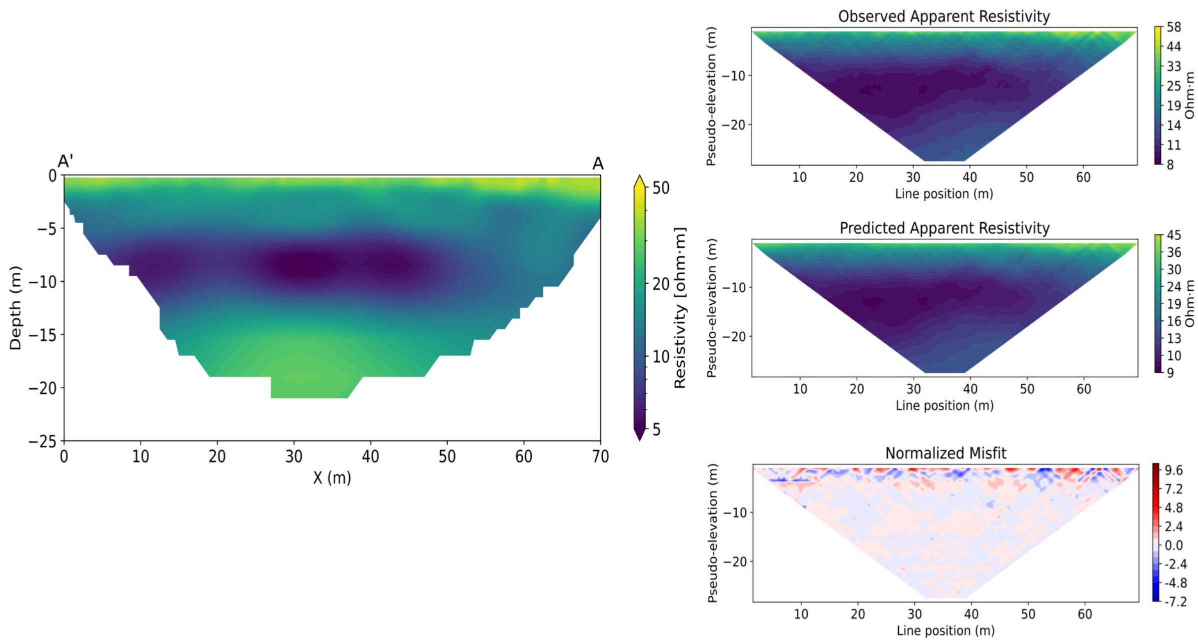

Figure A.2: Resistivity profile obtained from ERT inversion of Dike 2, Line 1. The final RMS was 1.73

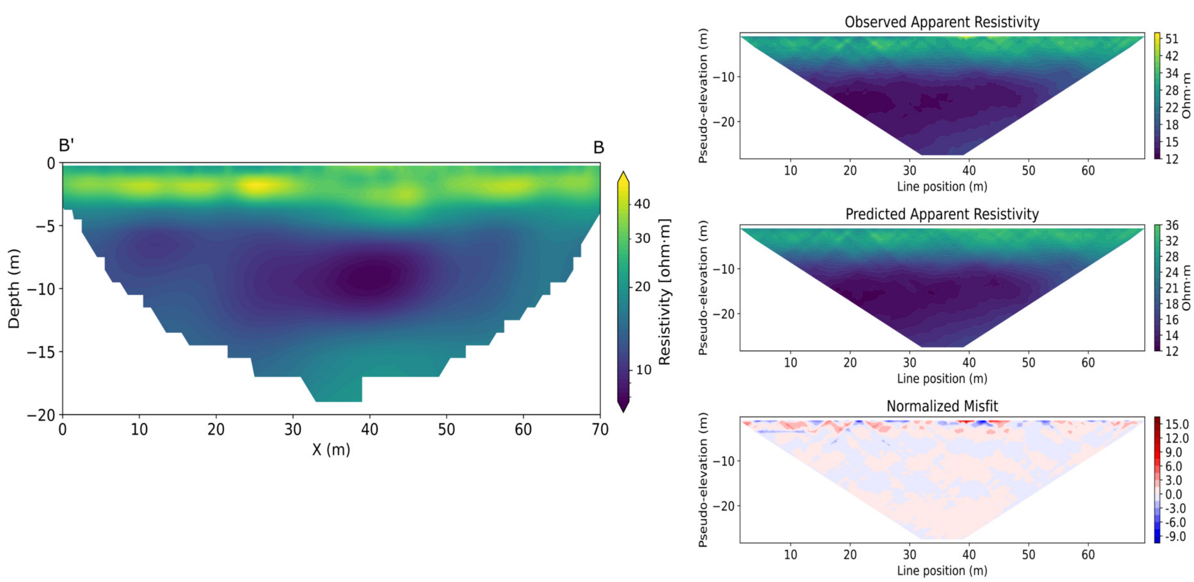

Figure A.3: Resistivity profile obtained from ERT inversion of Dike 2, Line 2. The final RMS was 1.92

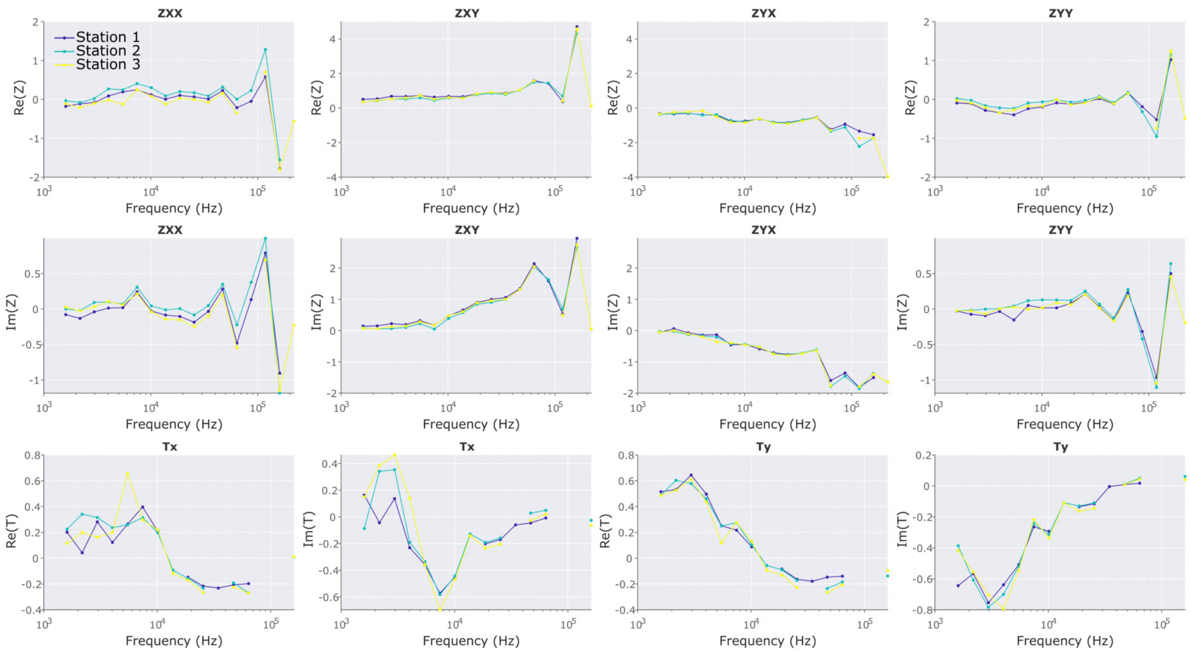

Figure A.4: Transfer function elements for all stations of Dike 2. Different colors represent data from different stations.

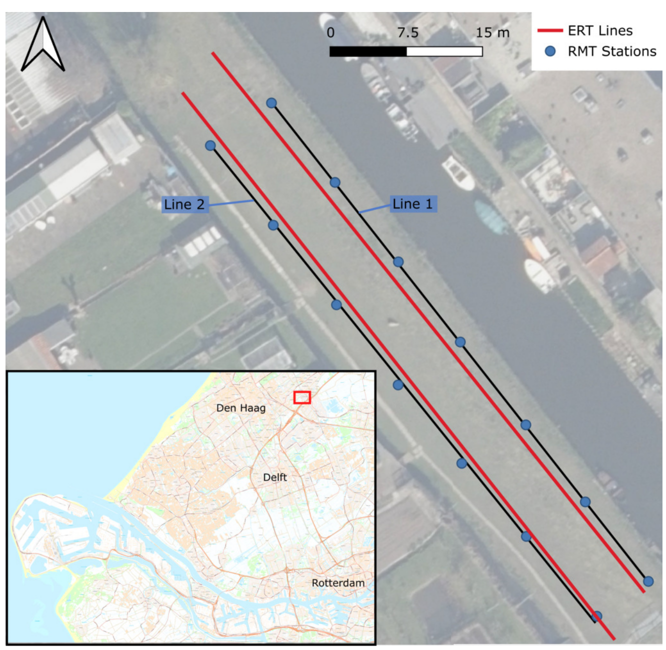

Figure A.5: Survey Layout of Dike 4 with the two RMT lines and two ERT lines.

Table A.2: Overview of acquisition parameters for Dike 4.

Parameter	**Value**
Location	52° 4’28.04”N, 4°23’13.55”E
Date	12 March 2026
Number of measurements	14 RMT stations
RMT Electrode spacing	10 m
RMT sampling frequency	524,288 Hz
Weather conditions	Sunny
Instruments	2 × ADU-08e (Metronix)
Measurement sequence	RMT lines 1 and 2 simultaneous

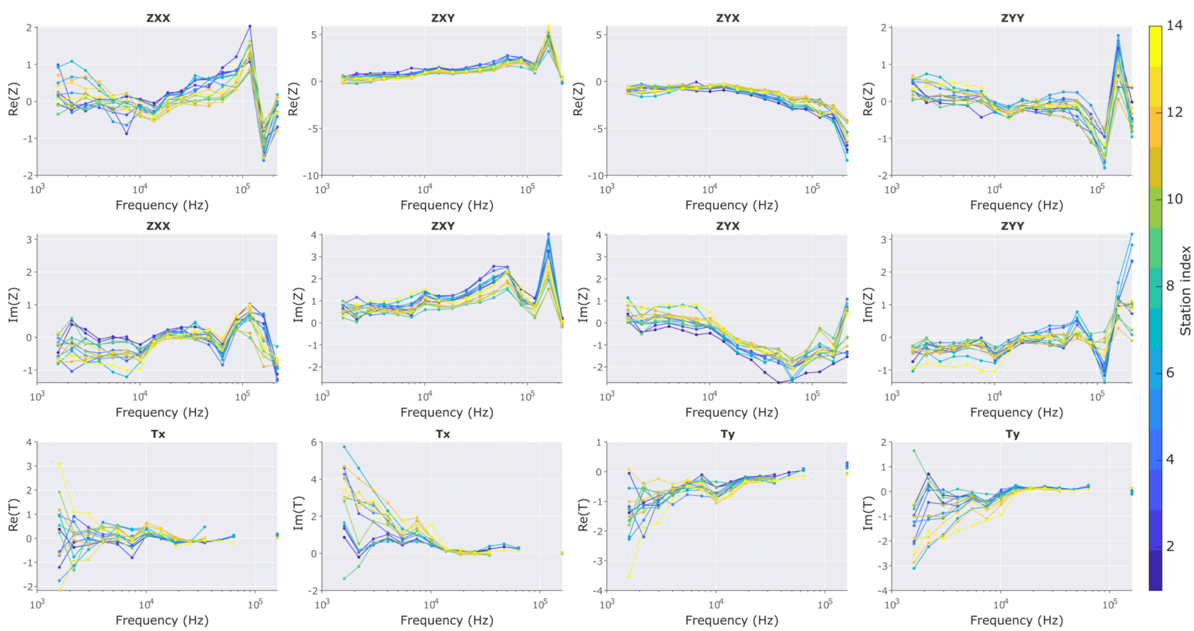

Figure A.6: Transfer function elements for all stations of Dike 4. Different colors represent data from different stations.

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Swiss Federal Institute of Technology Zurich

Eigenständigkeitserklärung

Die unterzeichnete Eigenständigkeitserklärung ist Bestandteil jeder während des Studiums verfassten schriftlichen Arbeit. Eine der folgenden zwei Optionen ist **in Absprache mit der verantwortlichen Betreuungsperson** verbindlich auszuwählen:

☐ Ich erkläre hiermit, dass ich die vorliegende Arbeit eigenverantwortlich verfasst habe, namentlich, dass mir niemand beim Verfassen der Arbeit geholfen hat. Davon ausgenommen sind sprachliche und inhaltliche Korrekturvorschläge der Betreuungsperson. Es wurden keine Technologien der generativen künstlichen Intelligenz[1] verwendet.

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Titel der Arbeit:

The Potential of Radiomagnetotellurics for Detecting Dike Instabilities

Verfasst von:
Bei Gruppenarbeiten sind die Namen aller Verfasserinnen und Verfasser erforderlich.

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Ich bestätige mit meiner Unterschrift:
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- Ich habe alle Methoden, Daten und Arbeitsabläufe wahrheitsgetreu und vollständig dokumentiert.
- Ich habe alle Personen erwähnt, welche die Arbeit wesentlich unterstützt haben.

Ich nehme zur Kenntnis, dass die Arbeit mit elektronischen Hilfsmitteln auf Eigenständigkeit überprüft werden kann.

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[1] Für weitere Informationen konsultieren Sie bitte die Webseiten der ETH Zürich, bspw. https://ethz.ch/de/die-eth-zuerich/lehre/ai-in-education.html und https://library.ethz.ch/forschen-und-publizieren/Wissenschaftliches-Schreiben-an-der-ETH-Zuerich.html (Änderungen vorbehalten).